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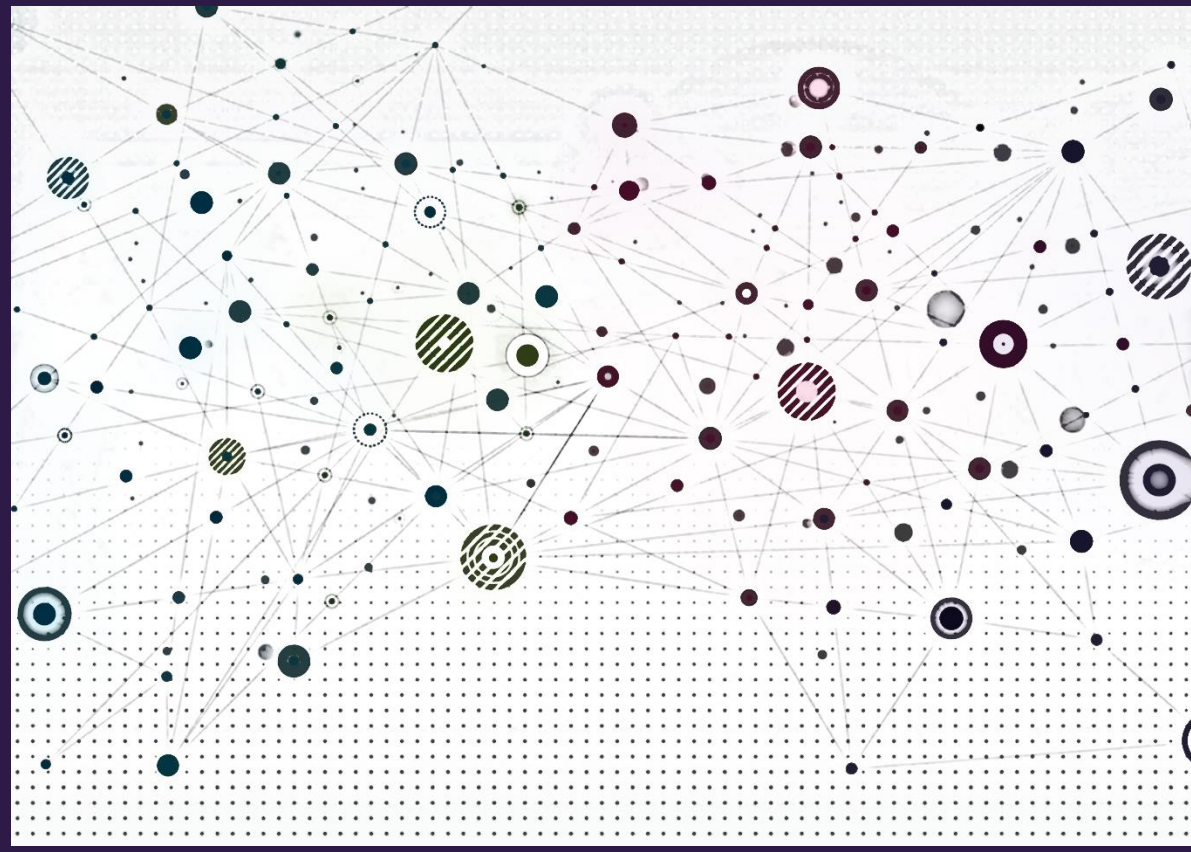
Reasoning on knowledge graphs



Overview

- Queries on knowledge graphs
- Predictive queries
- Predictive path queries with TransE
- Predictive conjunctive queries with Query2Box

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Queries on knowledge graphs

Queries (reasoning) on knowledge graphs

- How do we perform multi-hop (edge) reasoning over knowledge graphs (KGs)?

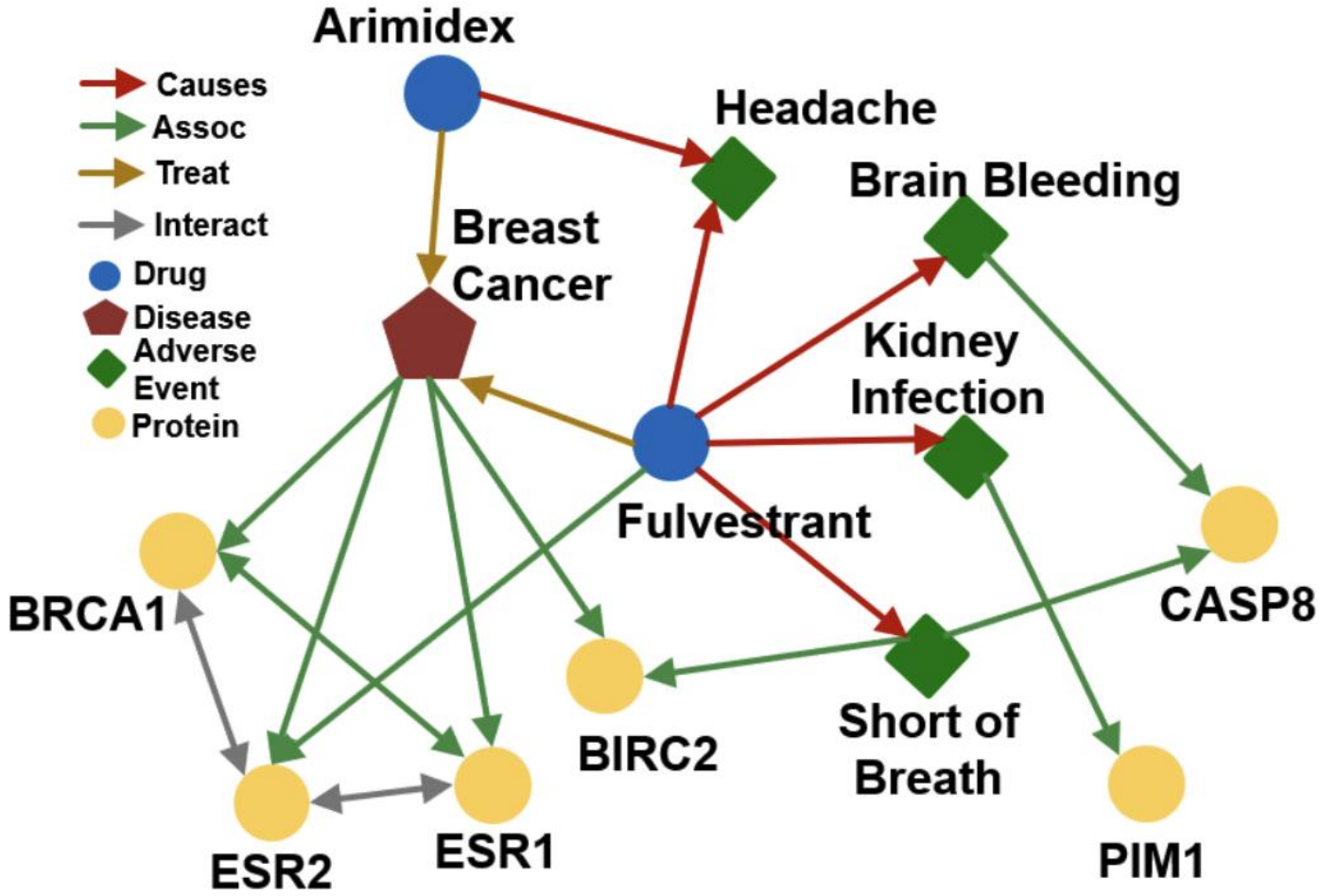


Image credit: Jure Leskovec, http://cs224w.stanford.edu

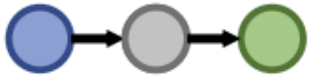
Types of multi-hop queries

Can we answer complex queries on a KG?

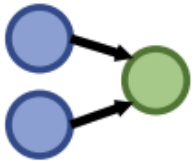
Query Types	Examples: Natural Language Question, Query
One-hop Queries	What adverse event is caused by Fulvestrant? (e:Fulvestrant, (r:Causes))
Path Queries	What protein is associated with the adverse event caused by Fulvestrant? (e:Fulvestrant, (r:Causes, r:Assoc))
Conjunctive Queries	What is the drug that treats breast cancer and caused headache? ((e:BreastCancer, (r:TreatedBy)), (e:Migraine, (r:CausedBy))

In this lecture, we only focus on answering queries on a KG!
The notation will be detailed next.

One-hop Queries



Path Queries

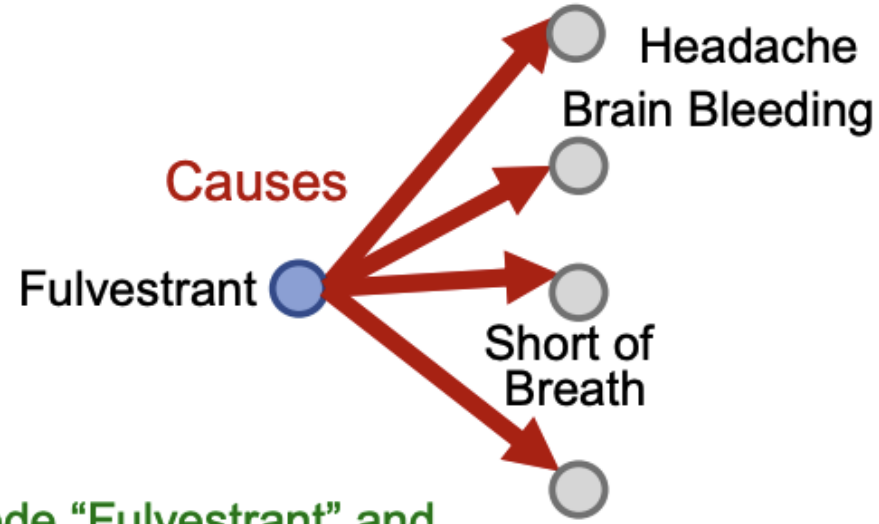


Conjunctive Queries

Image credit: Jure Leskovec, http://cs224w.stanford.edu

One-hop queries

- One-hope query: Is *t* an answer to the query *(h, r)*?
- Example: What side effects are caused by the drug Fulvestrant?
- Query: (e: Fulvestrant, (r:Causes))



Start from the anchor node "Fulvestrant" and traverse the KG by the relation "Causes", we reach entities {"Brain Bleeding", "Short of Breath", "Kidney Infection", "Headache"}.

Image credit: Jure Leskovec, http://cs224w.stanford.edu

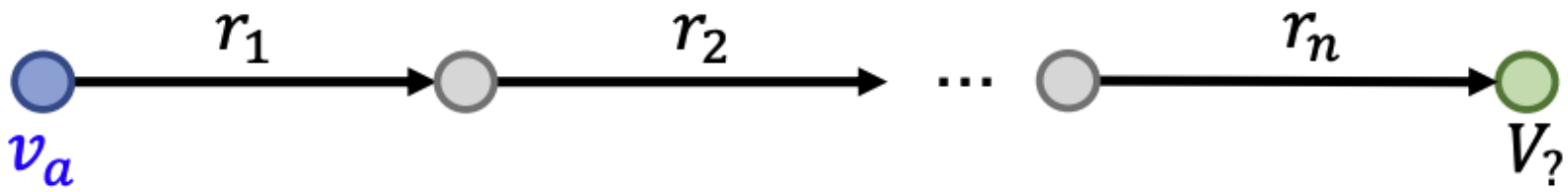
Path queries

- Generalize one-hop queries to path queries by adding more relations to the path
- An *n*-hop path query, q, can be represented by

$$q = (v_a, (r_1, \cdots, r_n))$$

  - v_a is an *anchor* entity
  - Let answers to q in graph G be denoted by $[\![q]\!]_G$

- Query plan for q is a chain (as are all path queries)



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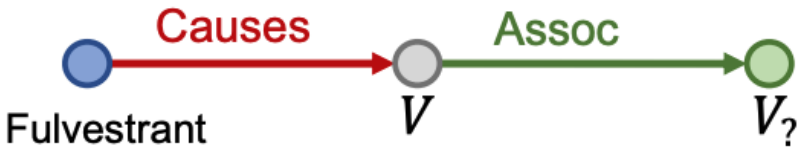
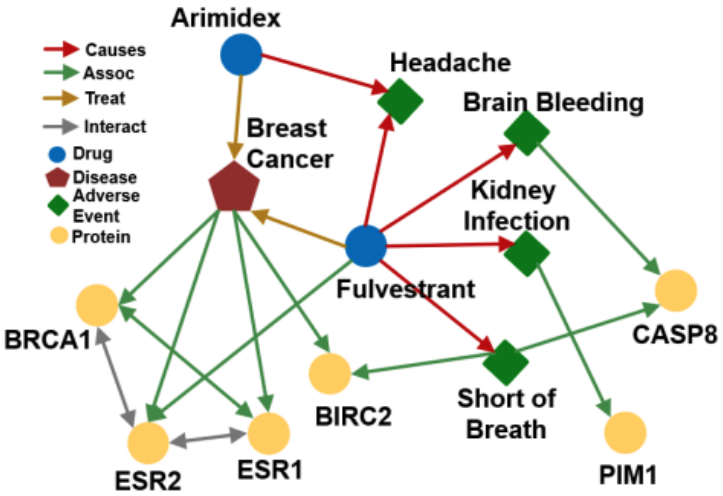
Path queries

Question: *What proteins are associated with adverse events caused by Fulvestrant?*

- v_a is e:Fulvestrant
- (r_1, r_2) is (r:Causes, r:Assoc)
- **Query:** (e:Fulvestrant, (r:Causes, r:Assoc))





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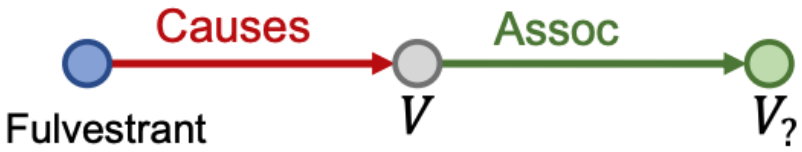
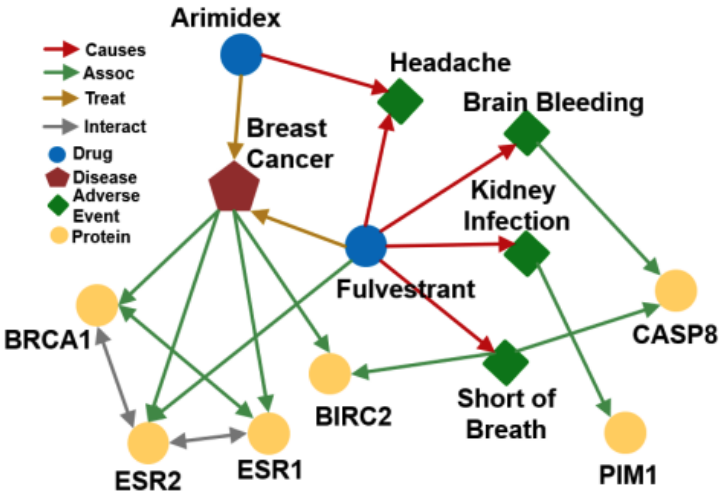
Path queries

Question: *What proteins are associated with adverse events caused by Fulvestrant?*

- **Query:** (e:Fulvestrant, (r:Causes, r:Assoc))
- Given a KG, how do we answer a path query?





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Traversing knowledge graphs for path queries

Question: *What proteins are associated with adverse events caused by Fulvestrant?*

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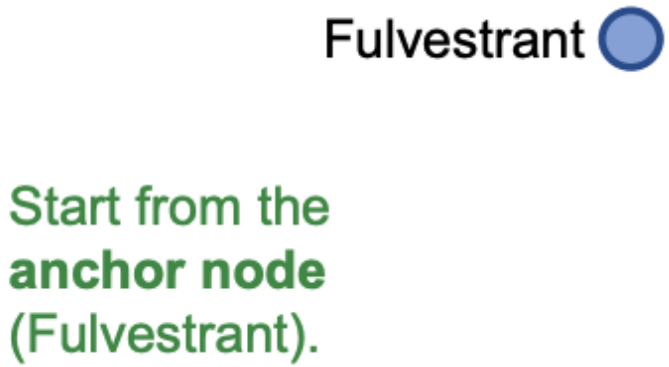

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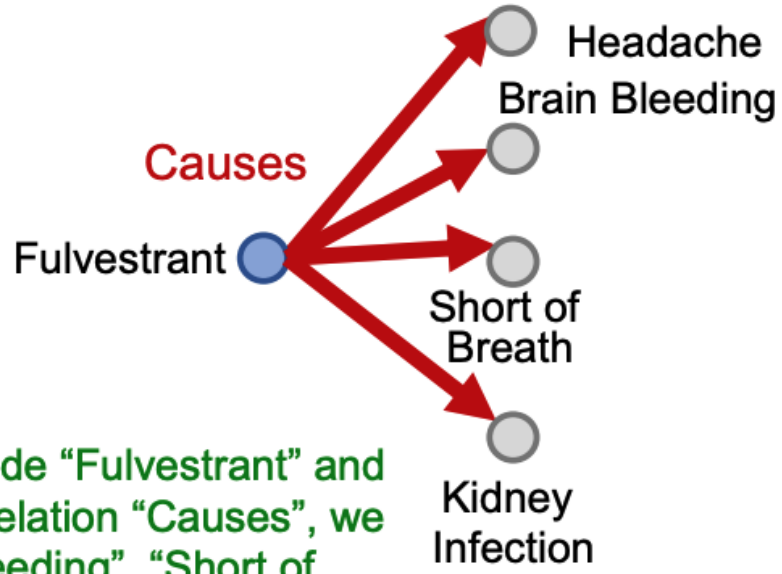

Start from the anchor node "Fulvestrant" and traverse the KG by the relation "Causes", we reach entities {"Brain Bleeding", "Short of Breath", "Kidney Infection", "Headache"}.

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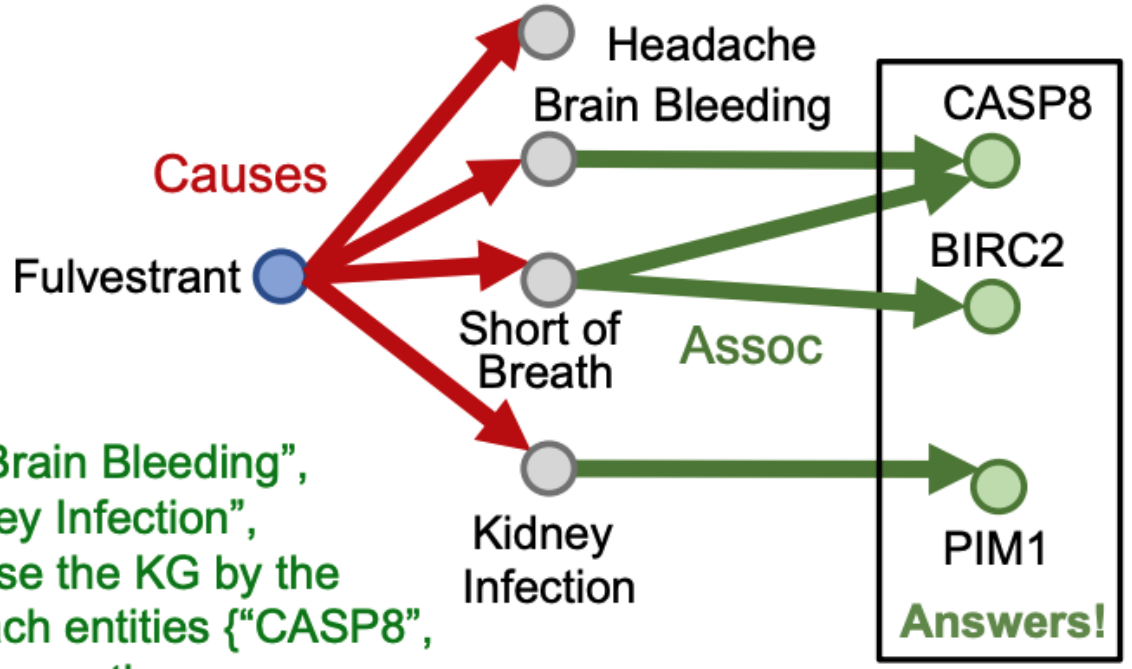


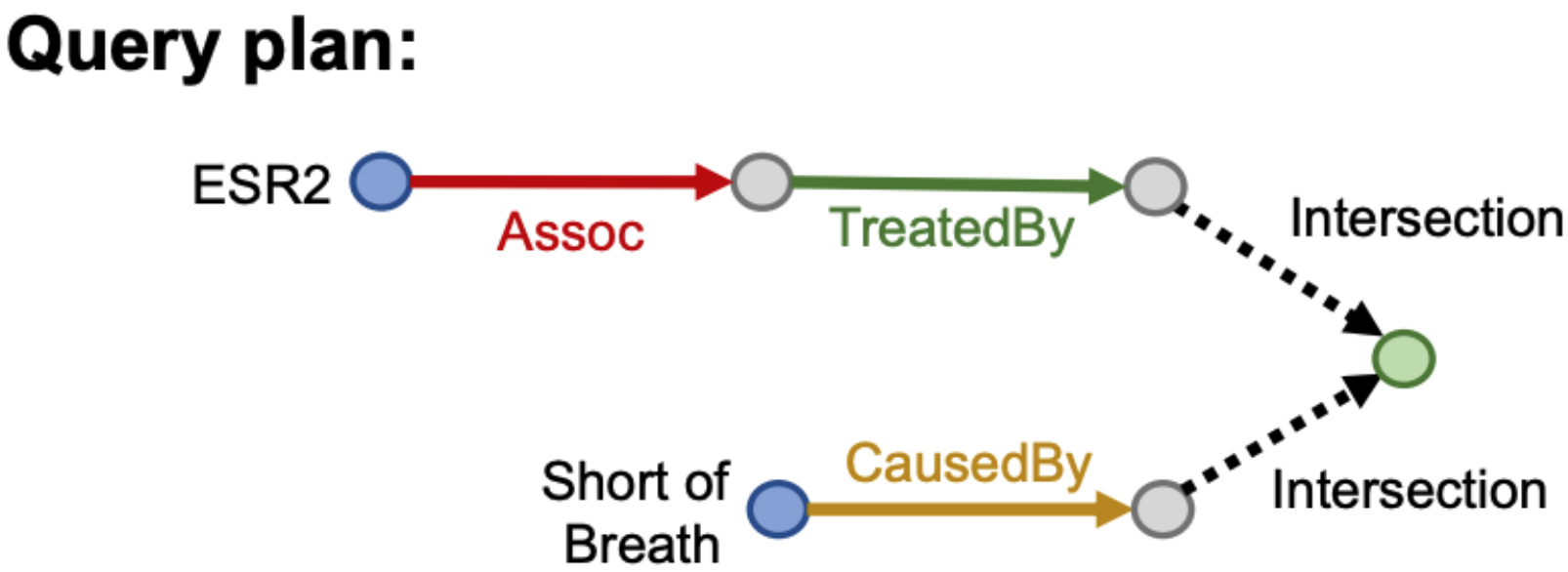
Start from the nodes {"Brain Bleeding", "Short of Breath", "Kidney Infection", "Headache"} and traverse the KG by the relation "Assoc", we reach entities {"CASP8", "BIRC2", "PIM1"}. These are the answers.

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Conjunctive queries

Question: *What are drugs that cause shortness of breath and treat diseases associated with protein ESR2?*

- **Query:** [(e:ESR2, (r:Assoc, r:TreatedBy)), (e:Short of Breath, (r:CausedBy))]



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Conjunctive queries

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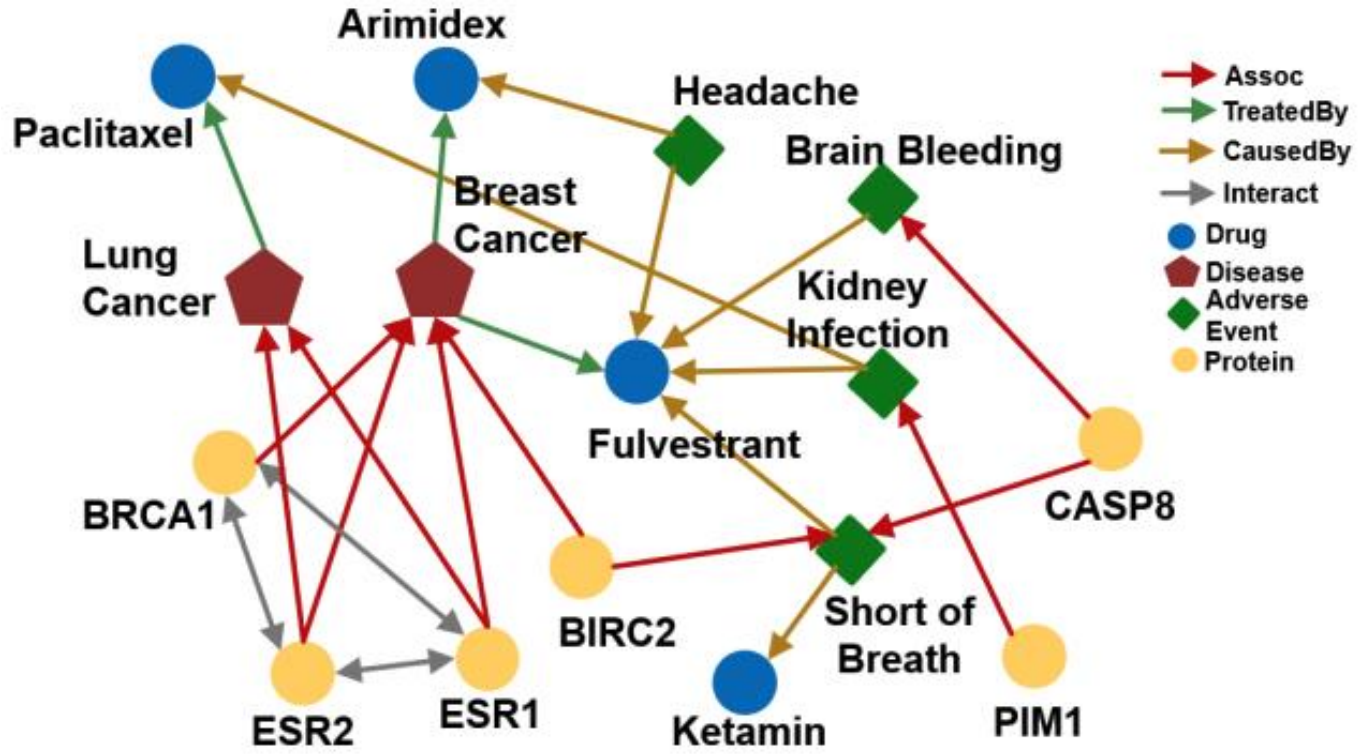
- **Query:** [(e:ESR2, (r:Assoc, r:TreatedBy)), (e:Short of Breath, (r:CausedBy))]
- How do we answer this question by KG traversal?



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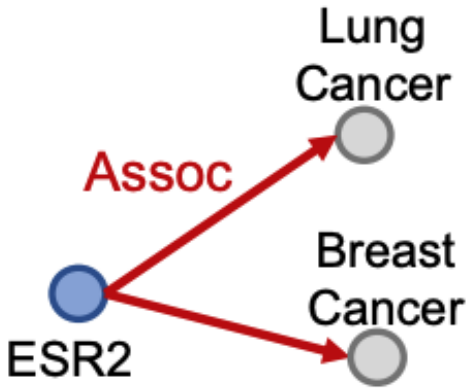
Traversing knowledge graphs for conjunctive queries

Question: *What are drugs that cause shortness of breath and treat diseases associated with protein ESR2?*

- **Query:** [(e:ESR2, (r:Assoc, r:TreatedBy)), (e:Short of Breath, (r:CausedBy))]
- Traverse KG from **anchor nodes**: ESR2 and Short of Breath



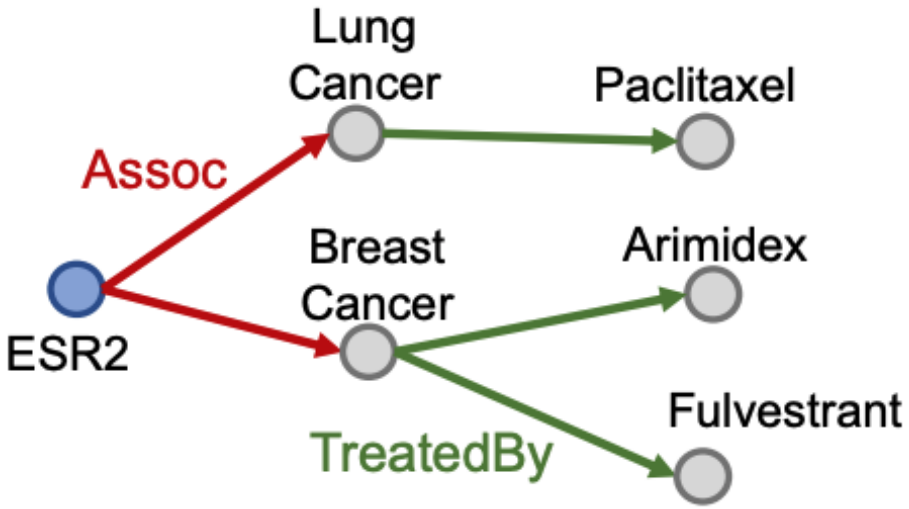
Traverse from the first anchor *ESR2* by relation *Assoc* to reach entities *{Lung Cancer, Breast Cancer}*

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Traversing knowledge graphs for conjunctive queries

Question: *What are drugs that cause shortness of breath and treat diseases associated with protein ESR2?*

- **Query:** [(e:ESR2, (r:Assoc, r:TreatedBy)), (e:Short of Breath, (r:CausedBy))]
- Traverse KG from **anchor nodes**: ESR2 and Short of Breath



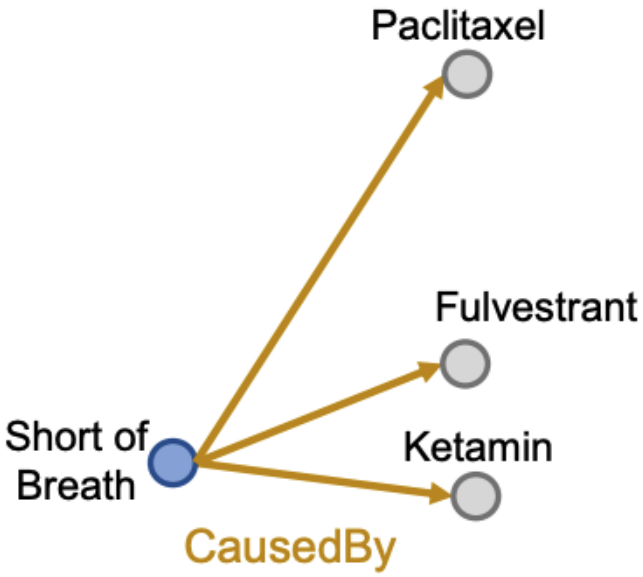
Traverse from the entities *{Lung Cancer, Breast Cancer}* by the relation *TreatedBy* to reach the drug entities *{Paclitaxel, Arimidex, Fulvestrant}*

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Traversing knowledge graphs for conjunctive queries

Question: *What are drugs that cause shortness of breath and treat diseases associated with protein ESR2?*

- **Query:** [(e:ESR2, (r:Assoc, r:TreatedBy)), (e:Short of Breath, (r:CausedBy))]
- Traverse KG from **anchor nodes**: ESR2 and Short of Breath



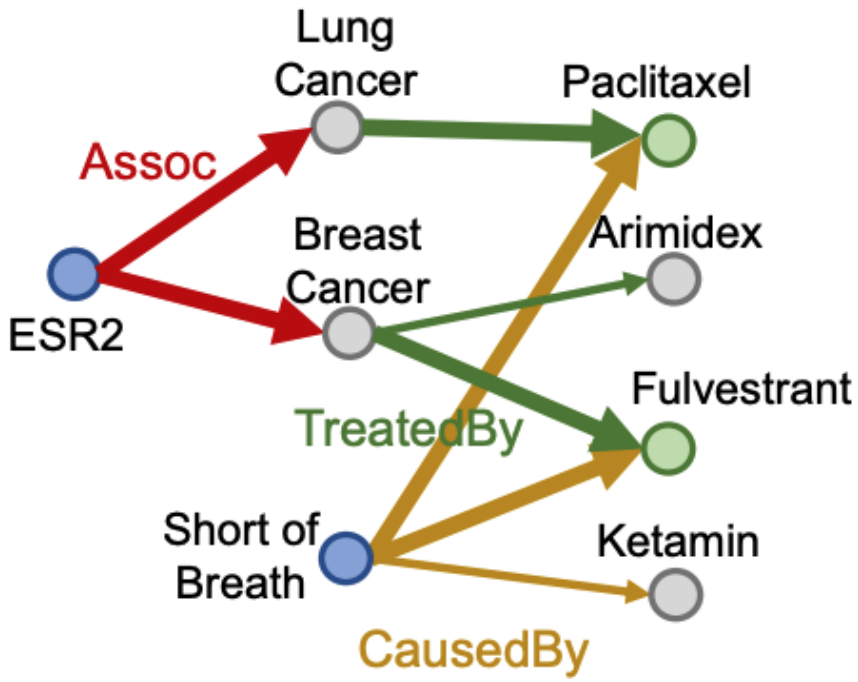
Traverse from the second anchor *Short of Breath* by the relation *CausedBy* to reach the set of entities *{Paclitaxel, Fulvestrant, Ketamin}*

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Traversing knowledge graphs for conjunctive queries

Question: *What are drugs that cause shortness of breath and treat diseases associated with protein ESR2?*

- **Query:** [(e:ESR2, (r:Assoc, r:TreatedBy)), (e:Short of Breath, (r:CausedBy))]
- Traverse KG from **anchor nodes**: ESR2 and Short of Breath



Take the intersection between the two drug sets to get the answer set *{Paclitaxel, Fulvestrant}*

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Predictive queries

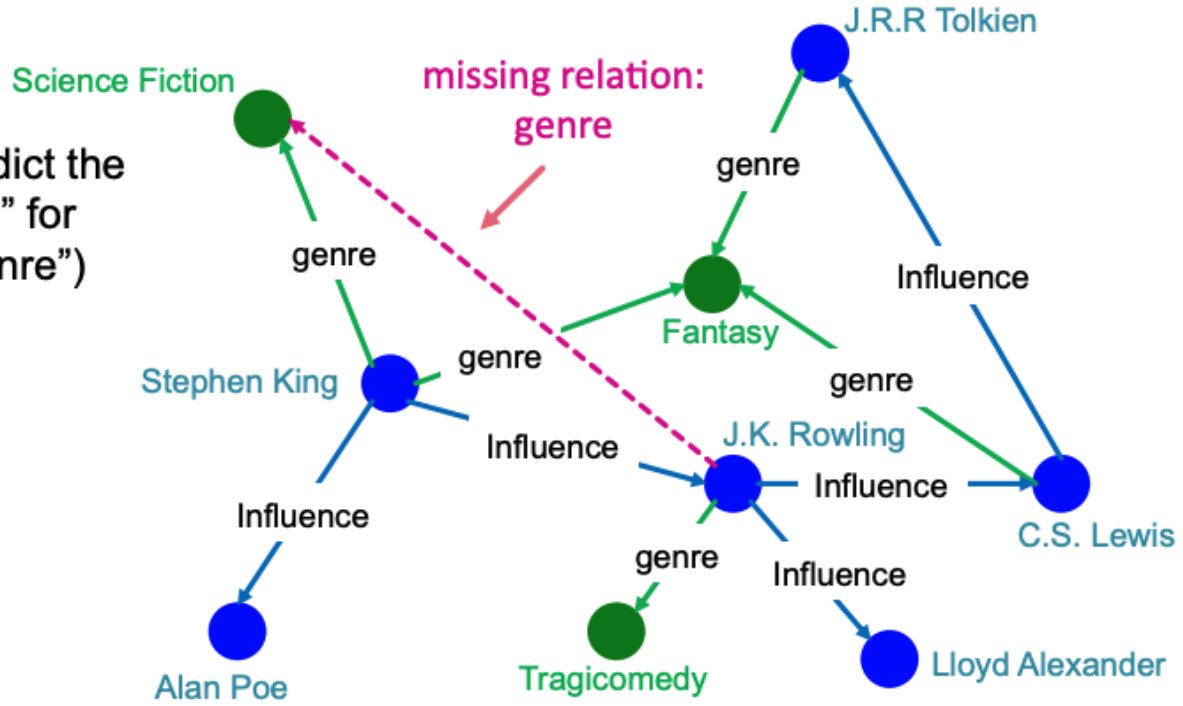
Answering queries on incomplete KGs

- Answering queries on KGs seems easy: just traverse the graph
- But KGs are incomplete
  - Many relations between entities are missing – for example, we lack knowledge of many associations among biomedical entities
  - Enumerating all facts requires non-trivial time and cost, so we shouldn't expect any KG to ever be complete
- Due to KG incompleteness, traversal alone will not identify all relations
  - Is this the best we can do?

Knowledge graph (KG) completion task

- Given a large KG, can we ”complete” it (i.e., add missing edges)
- Task is typically formulated as predicting tails (or consequents)
  - Consider relations in a directed KG represented as a tuple (head, relation, tail) where head (h) has relation (r) with tail (t)
  - Given (head, relation) predict missing tails

Example task: predict the tail “Science Fiction” for (“J.K. Rowling”, “genre”)



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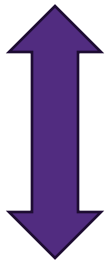
Predictive one-hop queries

- We can formulate KG completion problems as answering one-hop queries
- KG completion: Is the edge *(h, r, t)* in the KG?

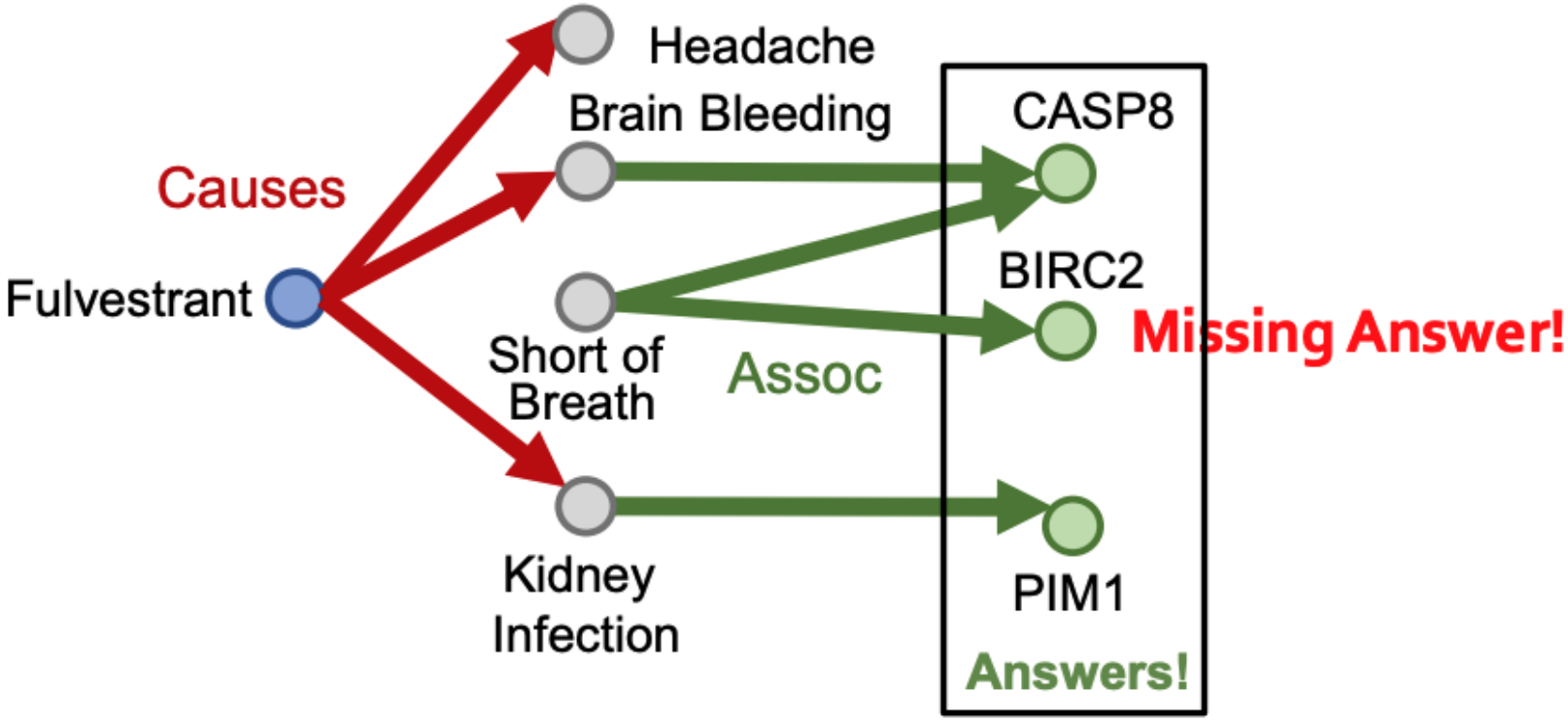


- One-hope query: Is *t* an answer to the query *(h, r)*?
  - Example: What side effects are caused by the drug Fulvestrant?

Predictive path queries?

Question: *What proteins are associated with adverse events caused by Fulvestrant?*

- **Query:** (e:Fulvestrant, (r:Causes, r:Assoc))



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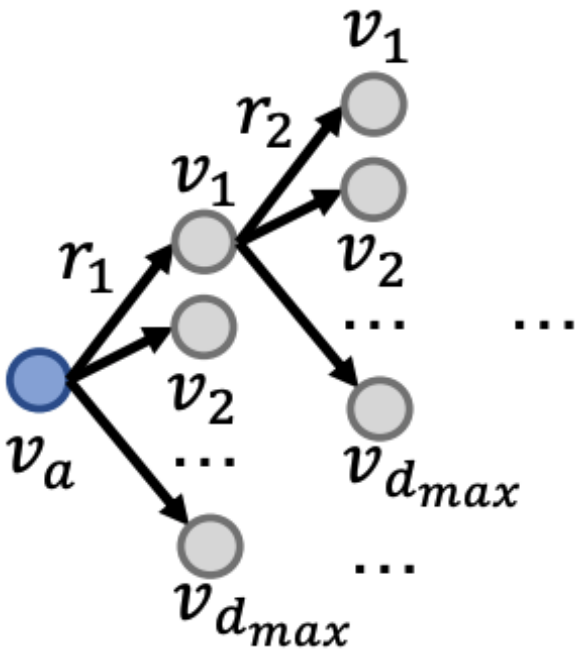
Predictive path queries

Can we first perform KG completion on the entire KG and then traverse the completed KG?

- No! The "completed" KG is a dense graph because most *(h, r, t)* triples will have some non-zero probability
- Time complexity of traversing a dense KG is $O(d_{max}^{L})$ exponential with respect to the path length L
- What else can we do?



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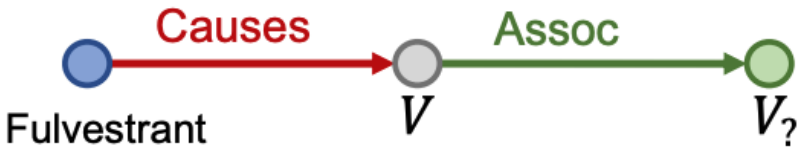
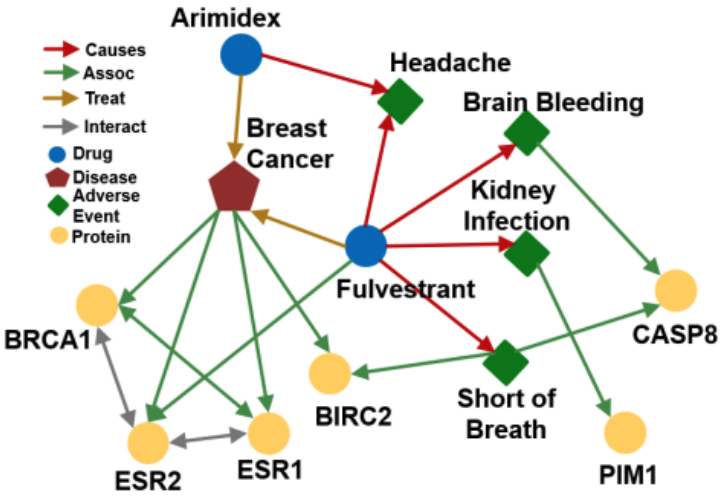
Predictive queries

- We need methods to answer path-based queries on an incomplete KG
  - Want to implicitly impute and account for missing relations (incompleteness)
  - Want to be able to answer arbitrary queries
- Generalization of the link prediction task





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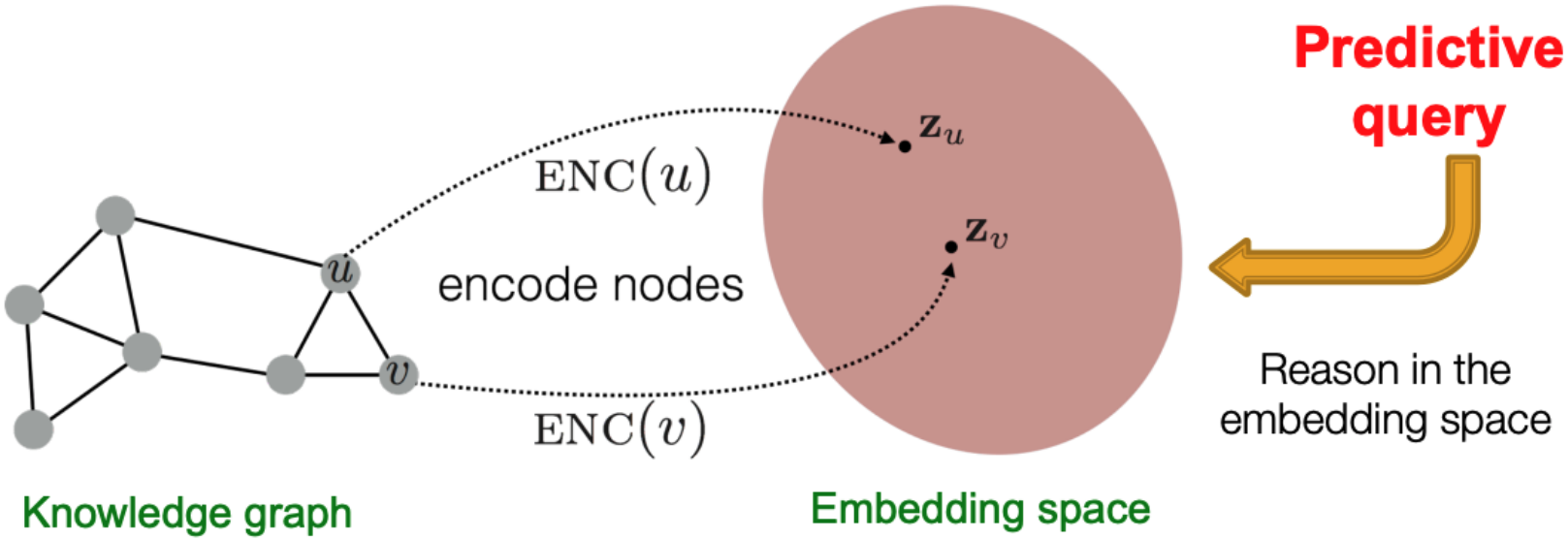
Embedding queries concept

Idea: Map queries into an embedding space and learn to reason in that space

- Embed query into a single **point** in the Euclidean space under the assumption that answer nodes are close to the query
- **Query2Box**: Embed query into a hyper-rectangle (**box**) in the Euclidean space under the assumption that answer nodes are in the box volume



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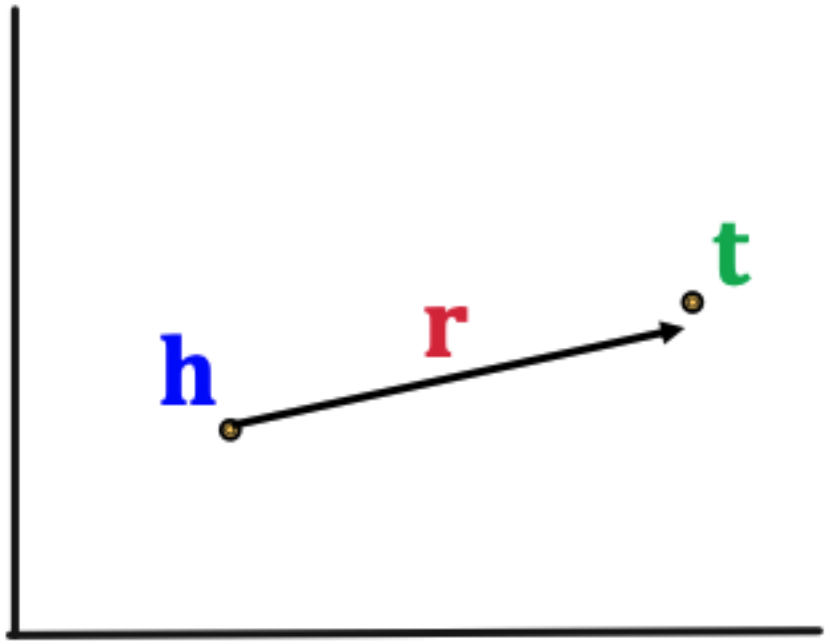
Predictive path queries with TransE

Trans(lation) Embedding (TransE)

- Let $\mathbf{h}, \mathbf{r}, \mathbf{t} \in \mathbb{R}^k$ be embeddings for the relation *(h, r, t)*
- **TransE:** $\mathbf{h} + \mathbf{r} \approx \mathbf{t}$ if the relation *(h, r, t)* exists, else $\mathbf{h} + \mathbf{r} \neq \mathbf{t}$
- Define the scoring function as $f_r(h, t) = -\|\mathbf{h} + \mathbf{r} - \mathbf{t}\|$
- Geometric interpretation as vectors will help us examine the ability of **TransE** to model different logic relations



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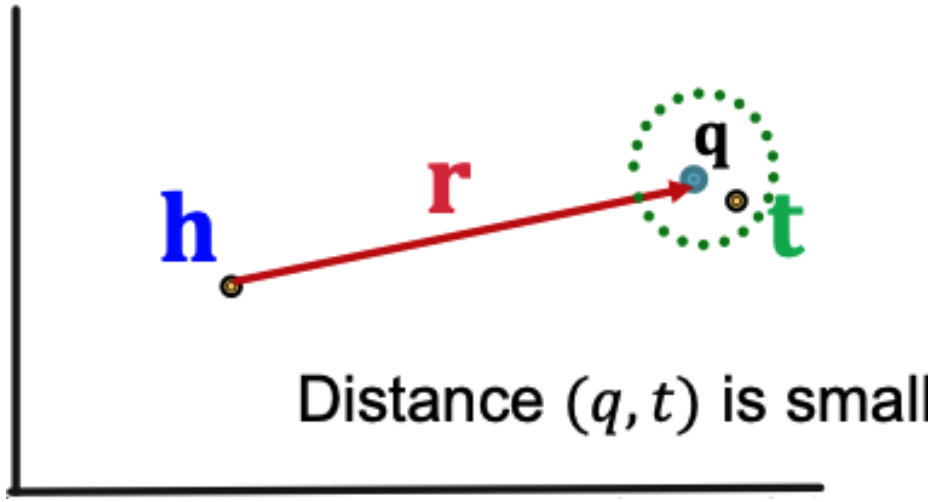
Predictive path queries with TransE

- Generalize TransE to path queries
- TransE: Translates $\mathbf{h}$ to $\mathbf{t}$ using $\mathbf{r}$ with score function $f_r(h,t) = -\|\mathbf{h}+\mathbf{r}-\mathbf{t}\|$
- Alternative interpretation
  - Query embedding: $\mathbf{q} = \mathbf{h} + \mathbf{r}$
  - Goal: move $\mathbf{q}$ close to the answer embedding $\mathbf{t}$

$$f_q(t) = -\|\mathbf{q}-\mathbf{t}\|$$



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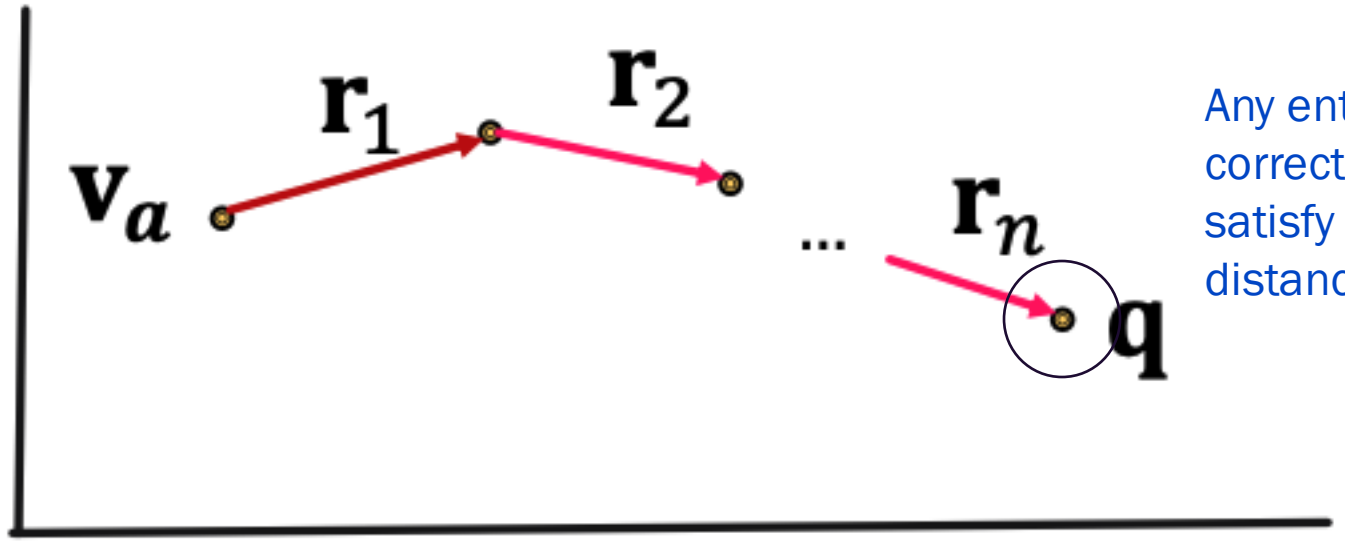
Predictive path queries with TransE

- Generalize TransE to path queries
- Given a path query : $q = (v_a, (r_1, \cdots, r_n))$, set the query embedding to

$$\mathbf{q} = \mathbf{v}_a + \mathbf{r}_1 + \cdots + \mathbf{r}_n$$

- The embedding process only involves vector addition that is independent of the number of entities in the KG



Any entity embeddings of the correct type (i.e., those that satisfy the query) within $\epsilon > 0$ distance of $\mathbf{q}$

Image credit: Jure Leskovec, http://cs224w.stanford.edu

An example of predictive path queries with TransE

- Question: *What proteins are associated with adverse events caused by Fulvestrant?*
- **Query:** (e:Fulvestrant, (r:Causes, r:Assoc))

Follow the query plan:

Image credit: Jure Leskovec, http://cs224w.stanford.edu

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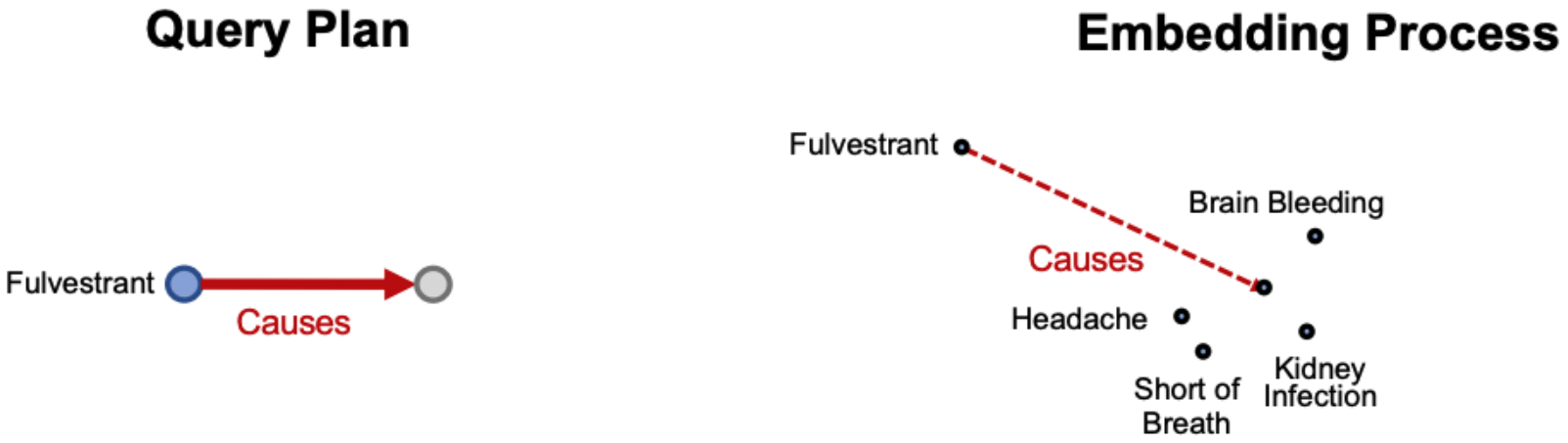

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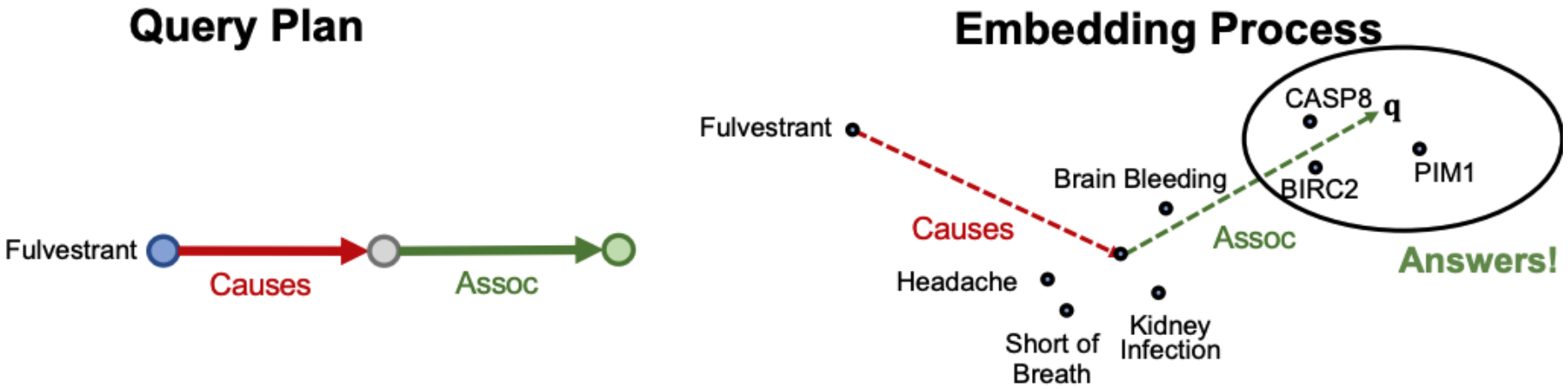

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Predictive path query training on TransE

- We can train a TransE model to optimize the knowledge graph completion objective (tail completion)
- Recall, TransE handles compositional relations, hence it naturally handles path queries over multiple hops
  - We get this for “free”!

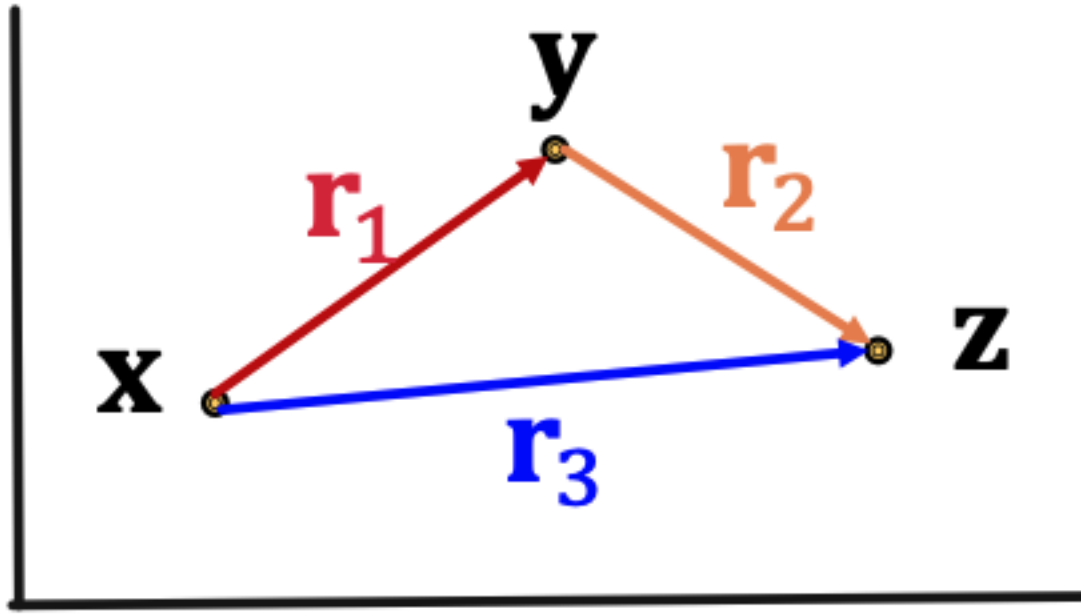


Image credit: Jure Leskovec, http://cs224w.stanford.edu

School of COMPUTING Clemson University

Predictive conjunctive queries with Query2Box

Predictive conjunctive queries

Question: *What are drugs that cause shortness of breath and treat diseases associated with protein ESR2?*

- **Query:** [(e:ESR2, (r:Assoc, r:TreatedBy)), (e:Short of Breath, (r:CausedBy))]
- Traverse KG from **anchor nodes**: ESR2 and Short of Breath

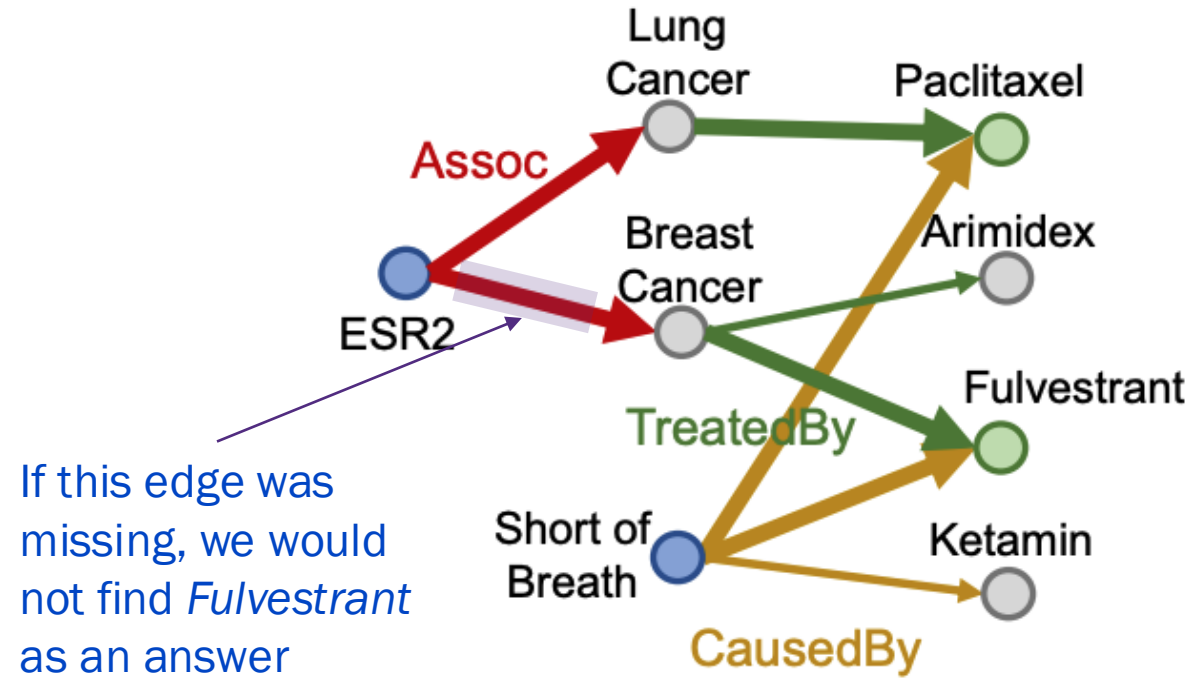


Image credit: Jure Leskovec, http://cs224w.stanford.edu

Predictive conjunctive queries

- How can we use embeddings to implicitly impute the missing relation *(ESR2, Assoc, Breast Cancer)*?
- Intuition: ESR2 interacts with BRCA1 and ESR1, both of which are associated with breast cancer.

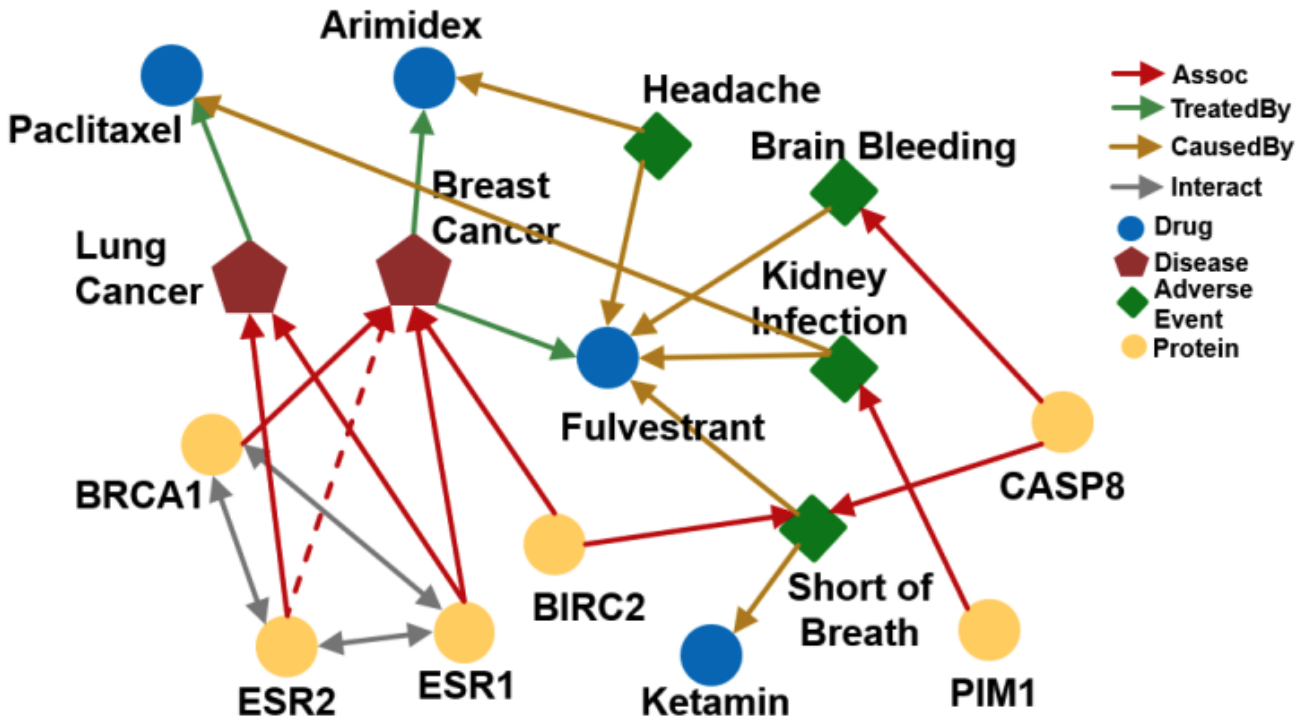


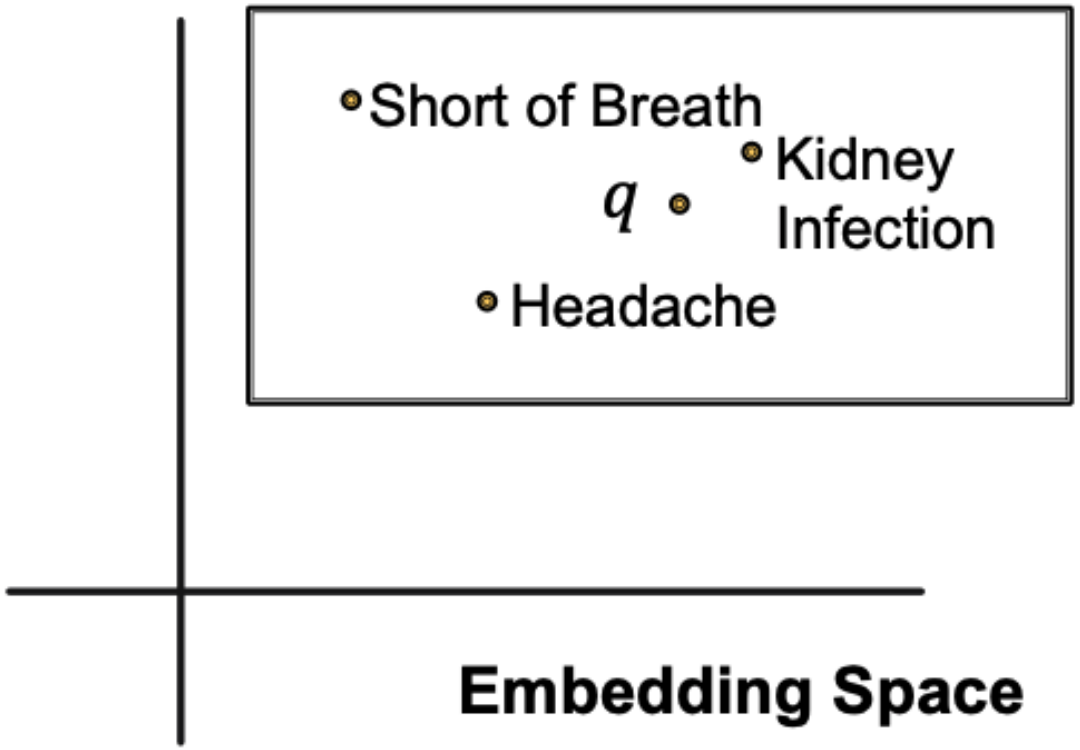
Image credit: Jure Leskovec, http://cs224w.stanford.edu

Box embeddings

- Embed queries, q, in a hyper-rectangle (box)

$$\mathbf{q} \equiv \{\mathbf{c_q}, \mathbf{o_q}\}$$

$\mathbf{q} \in \mathbb{R}^{2d}$ is the box
$\mathbf{c_q} \in \mathbb{R}^{d}$ center of box
$\mathbf{o_q} \in \mathbb{R}^{d}$ offset from center



Example: Embed the adverse events of Fulvestrant within a box that encloses all answer entities.

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Box embeddings

Some things we need to figure out:

- Entity embeddings (# parameters: $d|V|$)
  - Individual entities are *zero-volume* boxes (i.e., points)
- Relation embeddings (# parameters: $2d|R|$)
  - Each relation maps a box to a new box
- Intersection operator $\mathcal{J}$
  - Maps input boxes to output boxes
  - Models intersection of boxes

d: embedding dimension
$|V|$: number of entities (nodes)
$|R|$: number of relations

Embed conjunctive queries

Embed queries in vector space: *What are drugs that cause shortness of breath and treat diseases associated with protein ESR2?*

- **Query:** [(e:ESR2, (r:Assoc, r:TreatedBy)), (e:Short of Breath, (r:CausedBy))]
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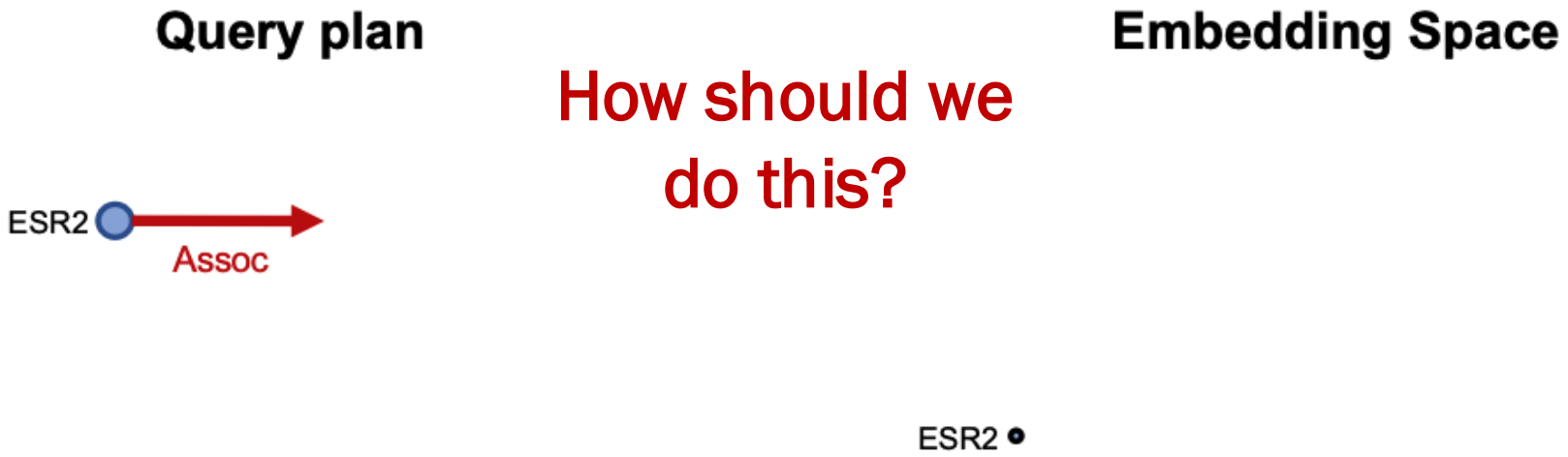


Image credit: Jure Leskovec, http://cs224w.stanford.edu

Projection operator

- Idea: Take the current box as input and use the relation embedding to project and expand the box
- Projection operator $\mathcal{P}: Box \times Relation \longrightarrow Box$

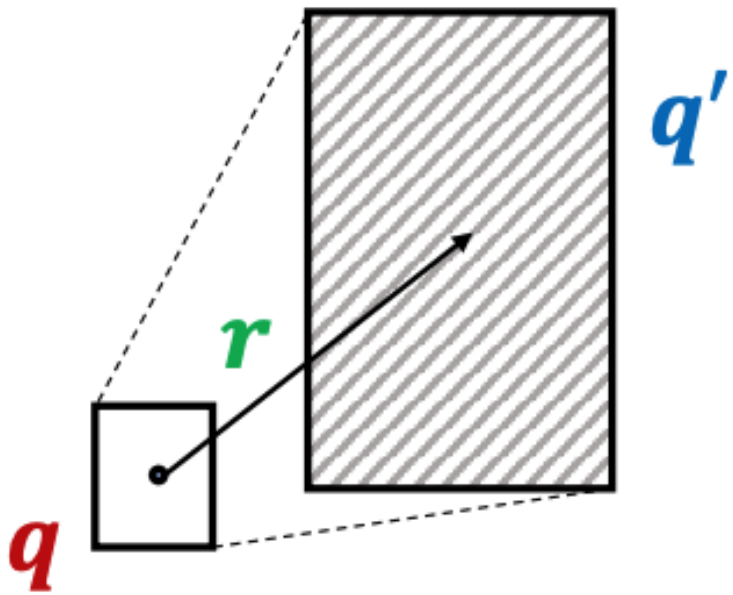
$\times$ means the projection operator is a mapping from an input box and relation to an output box

$\mathbf{c_q} \in \mathbb{R}^d$ center of box

$\mathbf{o_q} \in \mathbb{R}^d$ offset from center

$$\mathbf{c}_{q'} = \mathbf{c}_q + \mathbf{c}_r$$

$$\mathbf{o}_{q'} = \mathbf{o}_q + \mathbf{o}_r$$



Implementation: we don't actually learn an operator. Rather, we learn two (2) embeddings for every relation type in the graph, $\mathbf{c_r}$ and $\mathbf{o_r}$ that we use to translate and expand boxes

Image credit: Ren et al. 2020

Embed conjunctive queries

Embed queries in vector space: *What are drugs that cause shortness of breath and treat diseases associated with protein ESR2?*

- Traverse KG from **anchor nodes:** ESR2 and Short of Breath
- Use the projection operator to follow the relations in the query plan

Image credit: Ren et al. 2020

Embed conjunctive queries

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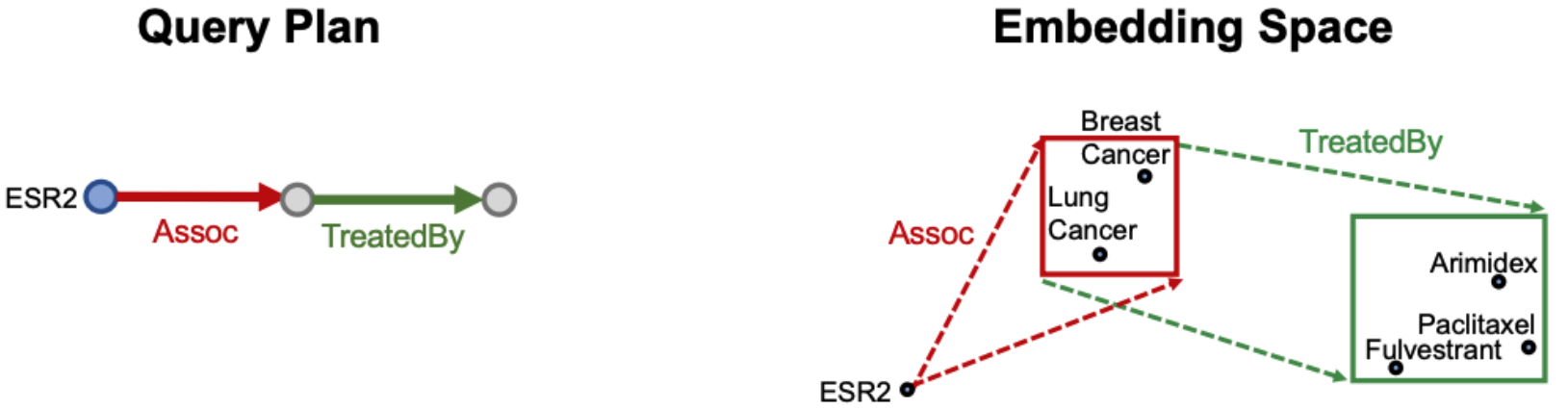

Image credit: Ren et al. 2020

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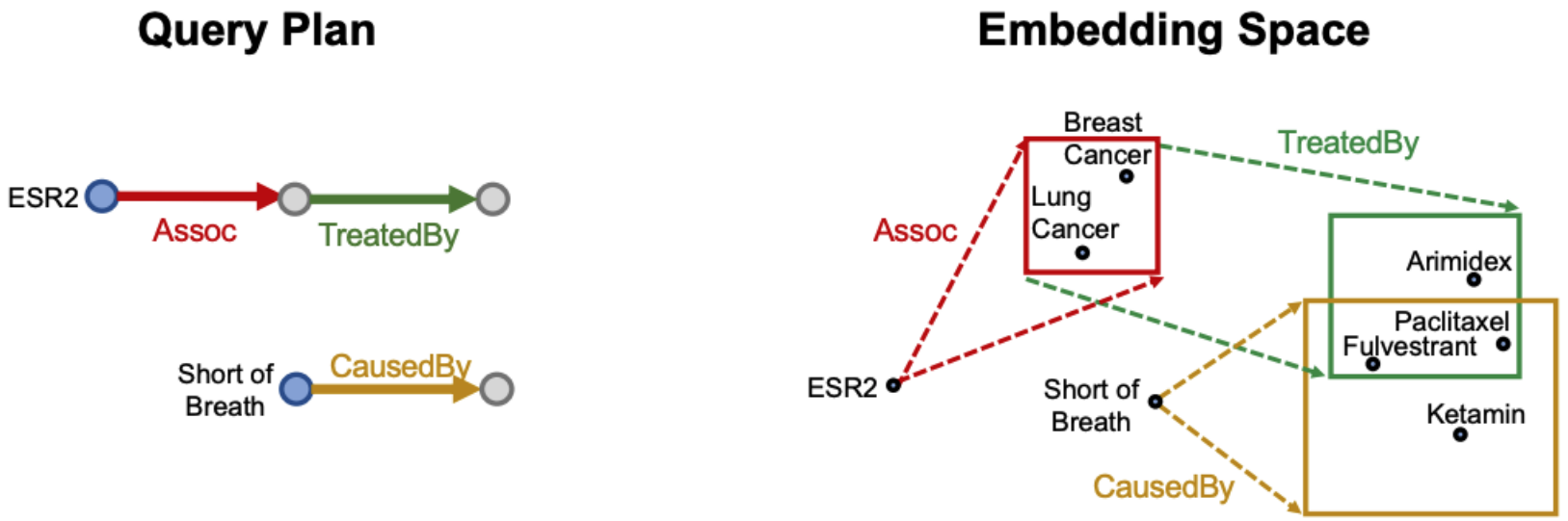
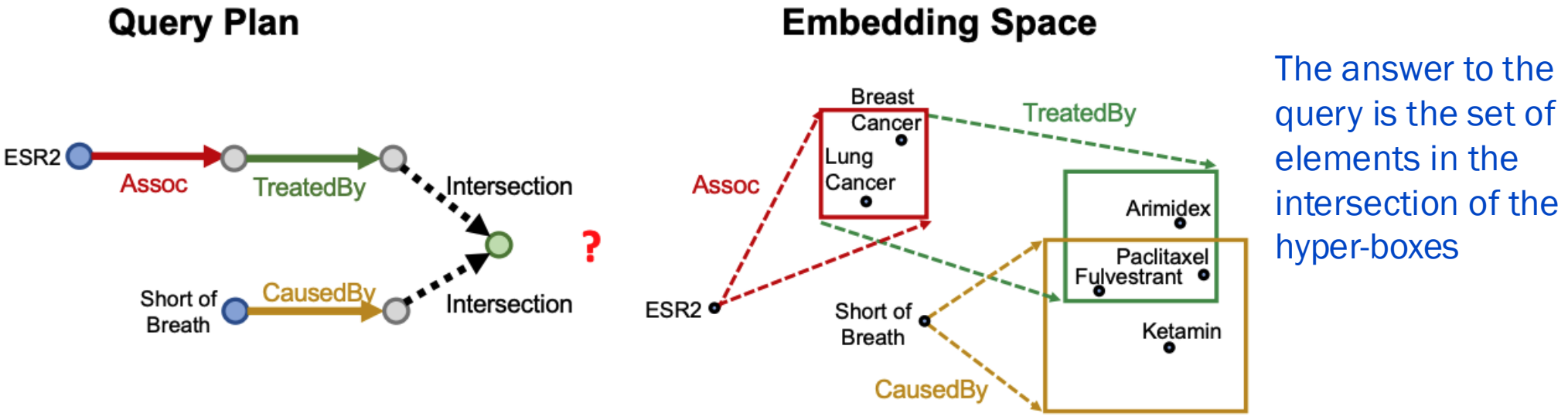


Image credit: Ren et al. 2020

Embed conjunctive queries

Embed queries in vector space: *What are drugs that cause shortness of breath and treat diseases associated with protein ESR2?*

- How do we take the intersection of the hyper-boxes?



The answer to the query is the set of elements in the intersection of the hyper-boxes

Image credit: Ren et al. 2020

Intersection operator

Intersection operator $\mathcal{J}: Box \times \cdots \times Box \longrightarrow Box$ is a mapping from multiple hyper-boxes to their intersection

- The center of the intersection should be “close” to the centers of the input boxes
- The offset (box size) should shrink since the size of the intersection should be smaller than the size of all the input sets

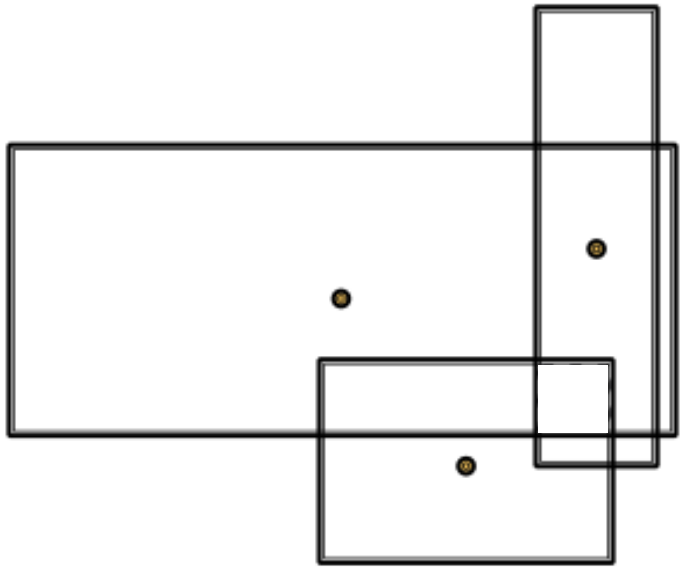


Image credit: Ren et al. 2020

Intersection operator

Intersection operator $\mathcal{J}: Box \times \cdots \times Box \longrightarrow Box$

$$\mathbf{c}_{\mathbf{q}_{int}} = \sum_i \mathbf{w}_i \odot \mathbf{c}_{\mathbf{q}_i}$$

MLP is a neural network with learnable parameters

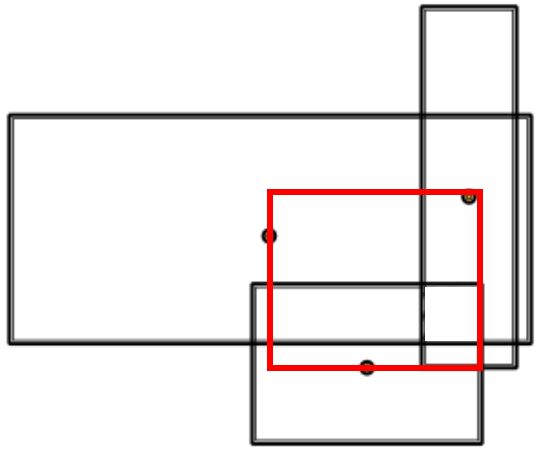
$$\mathbf{w}_i = \frac{exp[MLP(\mathbf{q}_i)]}{\sum_j exp[MLP(\mathbf{q}_j)]}$$

$\mathbf{q} \in \mathbb{R}^{2d}$ is the box

$\mathbf{c_q} \in \mathbb{R}^{d}$ center of box

$\mathbf{o_q} \in \mathbb{R}^{d}$ offset from center

$\odot$ is the Hadamard product (element wise multiplication



The center of the intersection is a weighted sum of the input box centers and is in the red region

$\mathbf{w}_i$ are attention scores over the individual input box centers

Image credit: Ren et al. 2020

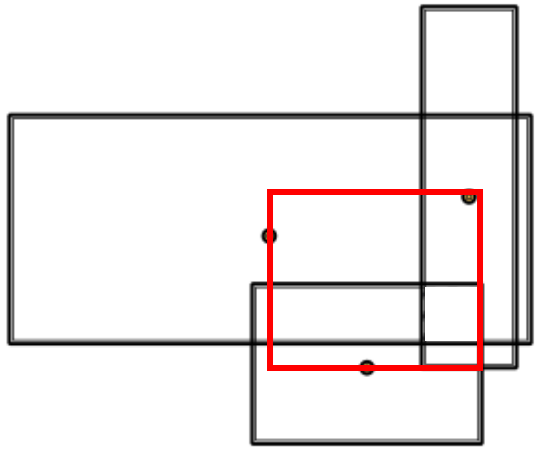
Intersection operator

Intersection operator $\mathcal{J}: Box \times \cdots \times Box \longrightarrow Box$

$$\mathbf{o}_{q_{int}} = \min_j \{\mathbf{o}_{\mathbf{q}_j}\} \odot \sigma[f_{\mathbf{o}}(\{\mathbf{q}_1, \cdots, \mathbf{q}_j\})]$$

σ sigmoid, guarantees shrinking of the box

$f_{\mathbf{o}}$ is a neural network with learnable parameters that operates on sets (e.g., DeepSets)



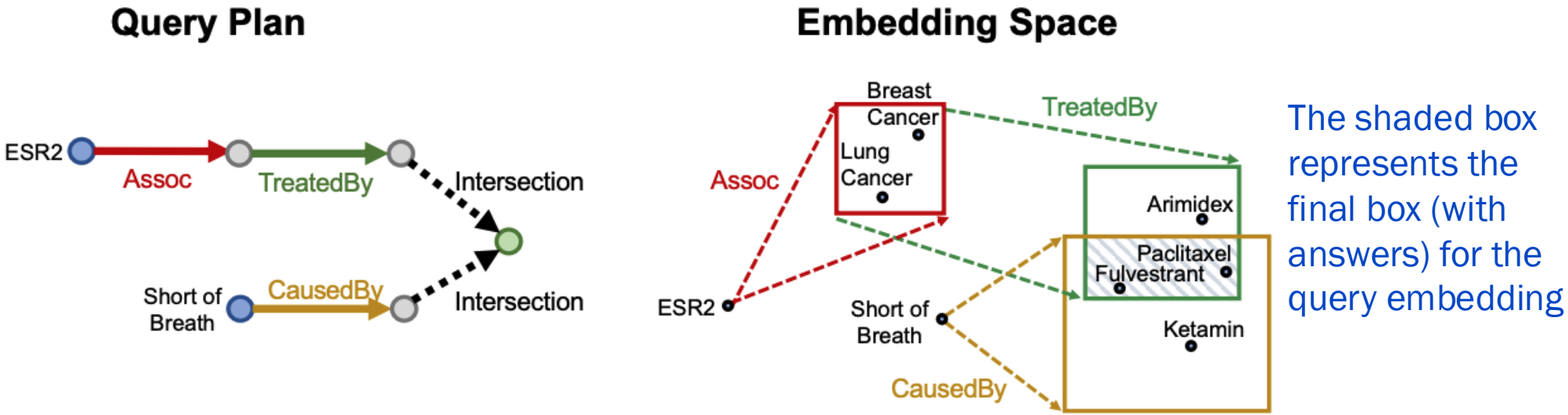
- The offset should be smaller than the offset of the input boxes but not necessarily equal to it
- Design a neural network, $f_{\mathbf{o}}$, to learn the appropriate scaling of the minimum input offset

Image credit: Ren et al. 2020

Embed conjunctive queries

Embed queries in vector space: *What are drugs that cause shortness of breath and treat diseases associated with protein ESR2?*

- Combine the projection and box intersection operators



The shaded box represents the final box (with answers) for the query embedding

Image credit: Ren et al. 2020

Distance to the query box

- Which entity embeddings are in the query box?
- Consider a query box $\mathbf{q}$ and an entity embedding $\mathbf{v}$
- Define the distance between $\mathbf{q}$ and $\mathbf{v}$ by

$$dist_{\text{box}}(\mathbf{v};\mathbf{q}) = dist_{\text{out}}(\mathbf{v};\mathbf{q}) + \alpha \cdot dist_{\text{in}}(\mathbf{v};\mathbf{q})$$

where $0 < \alpha < 1$ and

$$dist_{\text{out}}(\mathbf{v};\mathbf{q}) = \|max(\mathbf{v} - \mathbf{q}_{max}, \mathbf{0}) + max(\mathbf{q}_{min} - \mathbf{v}, \mathbf{0})\|_1$$

element wise *max* / *min*

$$dist_{\text{in}}(\mathbf{v};\mathbf{q}) = \left\|\mathbf{c}_\mathbf{q} - min\left(\mathbf{q}_{max}, max(\mathbf{q}_{min}, \mathbf{v})\right)\right\|_1$$

$$\|\mathbf{x}\|_1 = \sum_i |x_i|$$

where $\mathbf{q}_{max} = \mathbf{c}_\mathbf{q} + \mathbf{o}_\mathbf{q}$ and $\mathbf{q}_{min} = \mathbf{c}_\mathbf{q} - \mathbf{o}_\mathbf{q}$

Setting $0 < \alpha$ downweights distance in the box, indicating entities are "close enough" to the query center

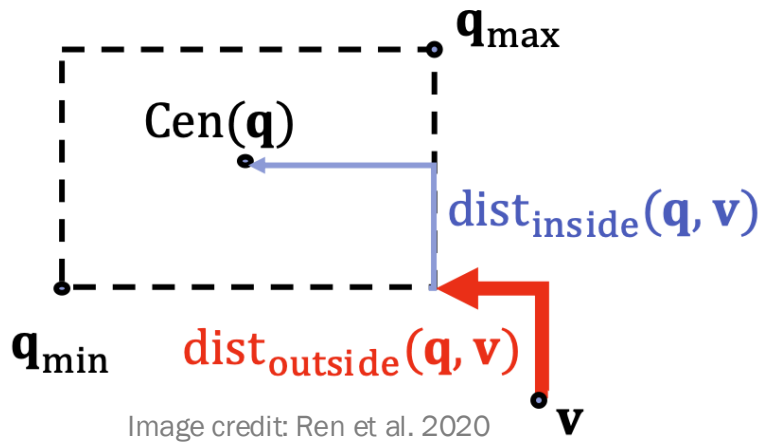


Image credit: Ren et al. 2020

Training Query2Box

- What are the trainable parameters?
  - Entity embeddings (# parameters: $d|V|$)
  - Relation embeddings (# parameters: $2d|R|$)
  - Intersection operator $\mathcal{J}$ which contains two neural networks
- Loss function

$$\mathcal{L} = -log\left[\sigma\left(\gamma - dist_{\text{box}}(\mathbf{v}; \mathbf{q})\right)\right] - \frac{1}{k}\sum_{i=1}^{k} log\left[\sigma(dist_{\text{box}}(\mathbf{v}_i'; \mathbf{q}) - \gamma)\right]$$

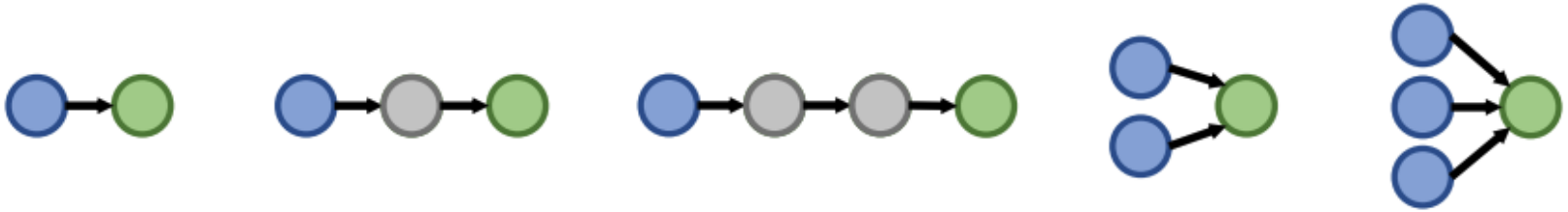
where γ is a scalar margin, $v \in [\![q]\!]$ is a positive entity (i.e., correct answer to the query, q) and $v_i' \notin [\![q]\!]$ is the *i*-th negative entity (non-answer to q)

Training Query2Box

- Training loop
  1. Sample a query from training data, *(q, v, v')*
  2. Use current embeddings and intersection operator to embed *q*
  3. Compute loss and update embedding and intersection operator weights
- How do we obtain training data and implement training?

Generating sample queries on a KG

Start with query templates that represent possible query plans in the KG of interest

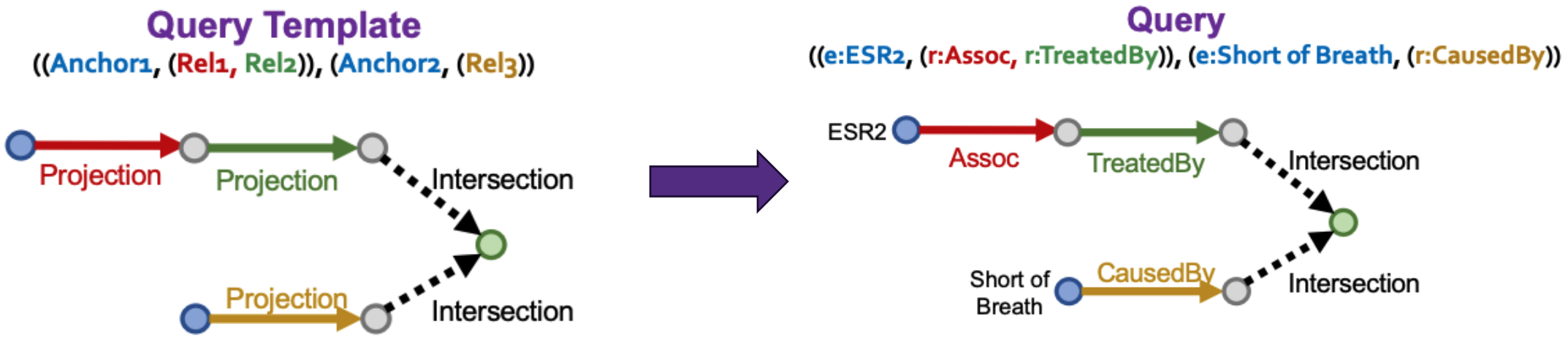


Different entity and relation types in the query templates as determined by the KG

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Generating sample queries on a KG

- Pick one of the query templates
- Generate sample queries by instantiating every variable with a concrete entity and relation from the KG
  - Example: instantiate Anhcor 1 with ESR2, Rel1 with Assoc
- How to do this systematically?



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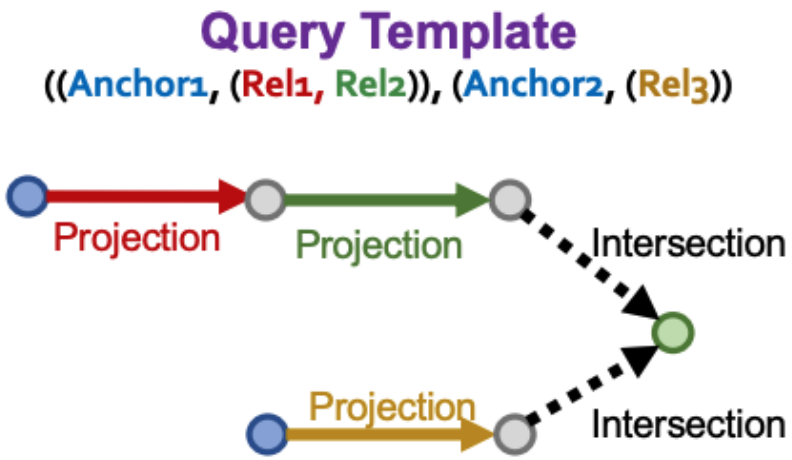
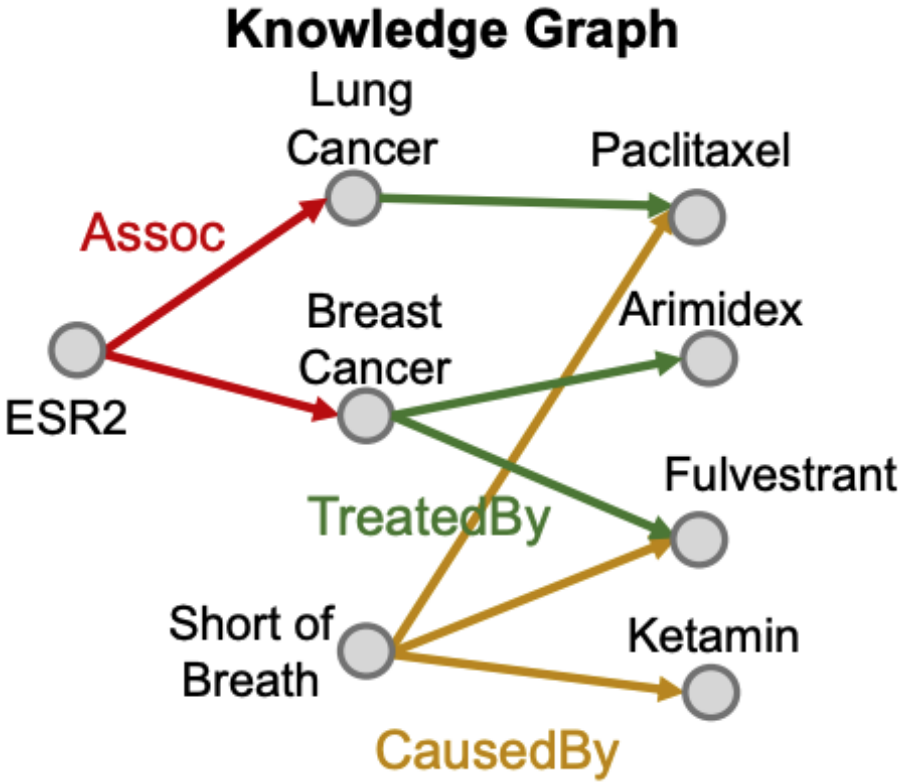
Generating sample queries on a KG

- How to do this systematically?



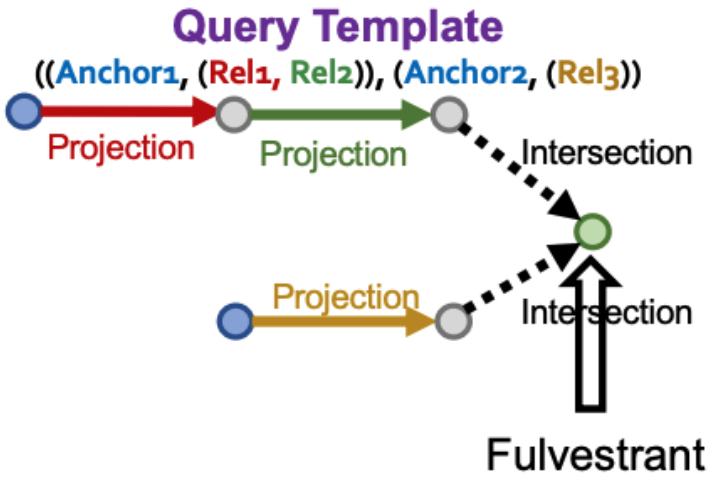


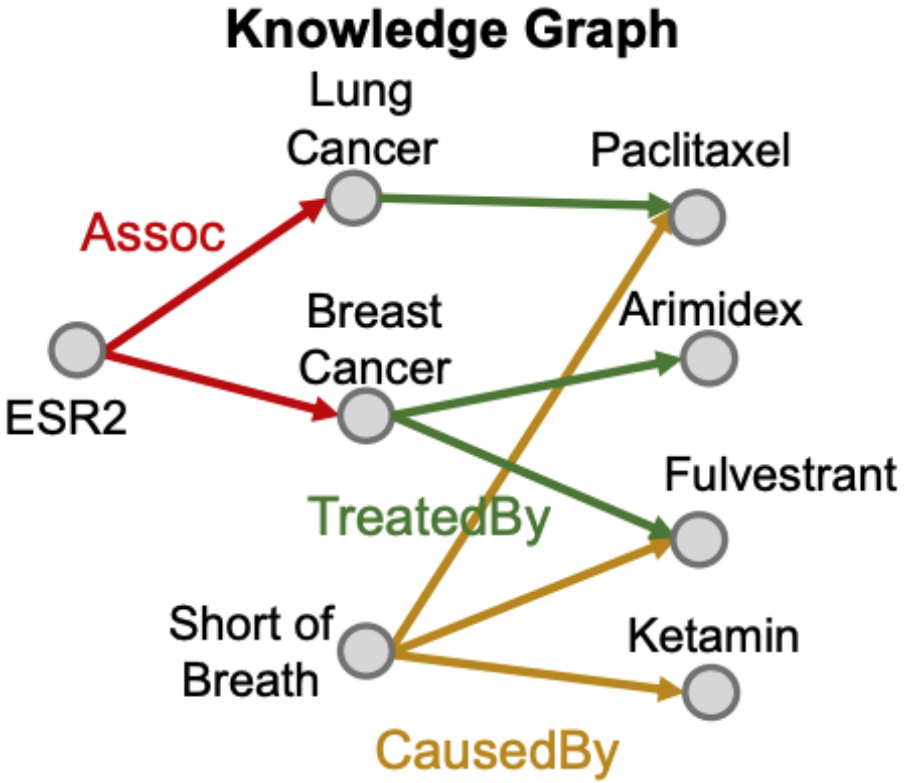
Overview: Start by instantiating an **answer node** of the query template and iteratively instantiate the other edges and nodes until we ground all anchor nodes

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Generating sample queries on a KG

- How to do this systematically?





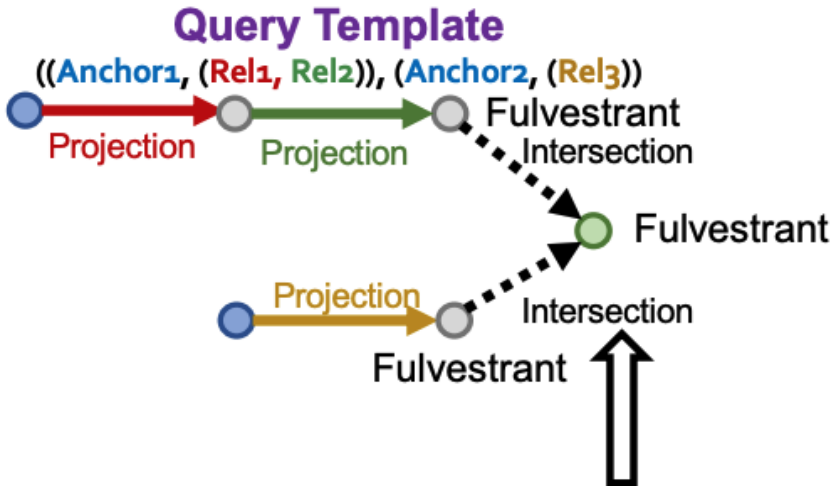
Start from the answer (root) of the query template. Why? As we will encounter intersections, we want to ensure the preceding paths include the final answer (see next slide)

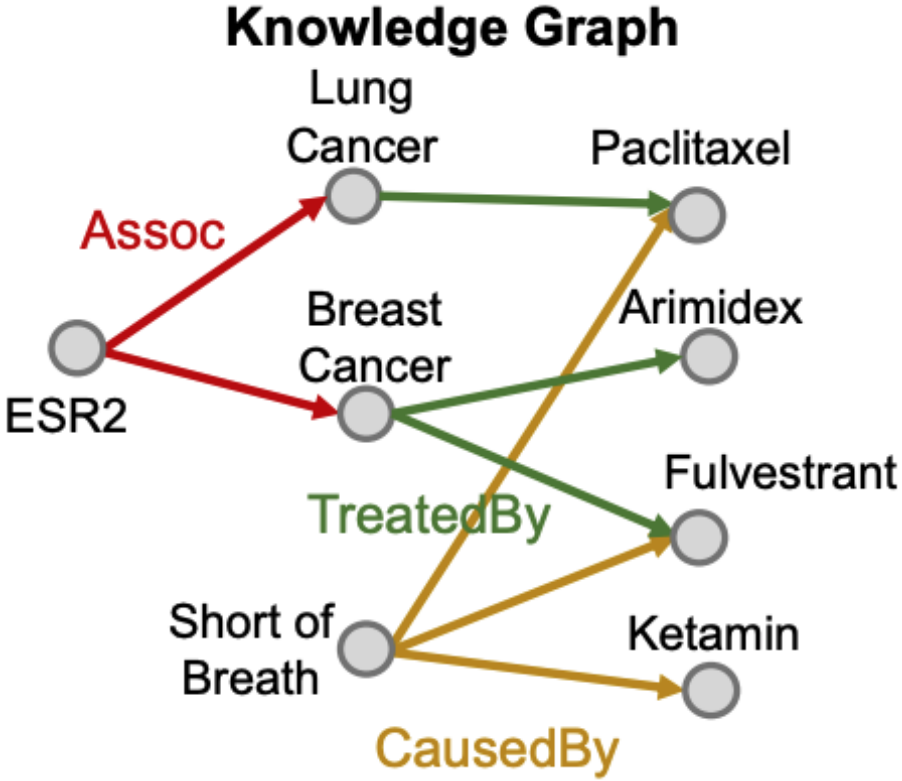
Randomly pick one entity from the KG as the root node – in this example, Fulvestrant

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Generating sample queries on a KG

- How to do this systematically?



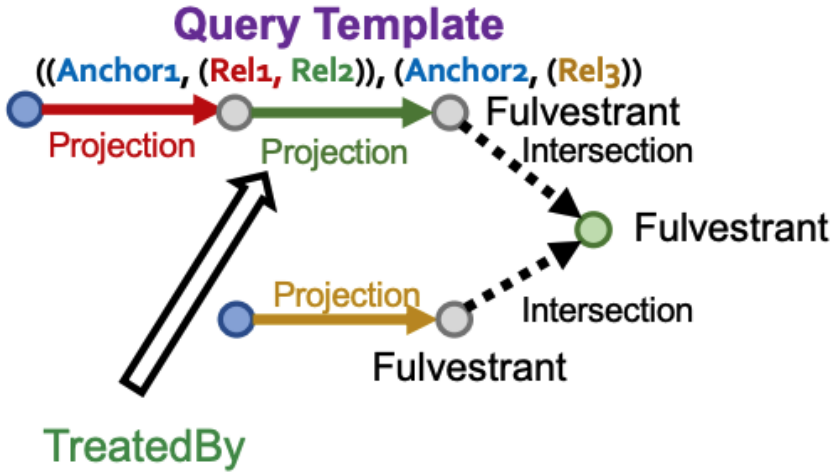


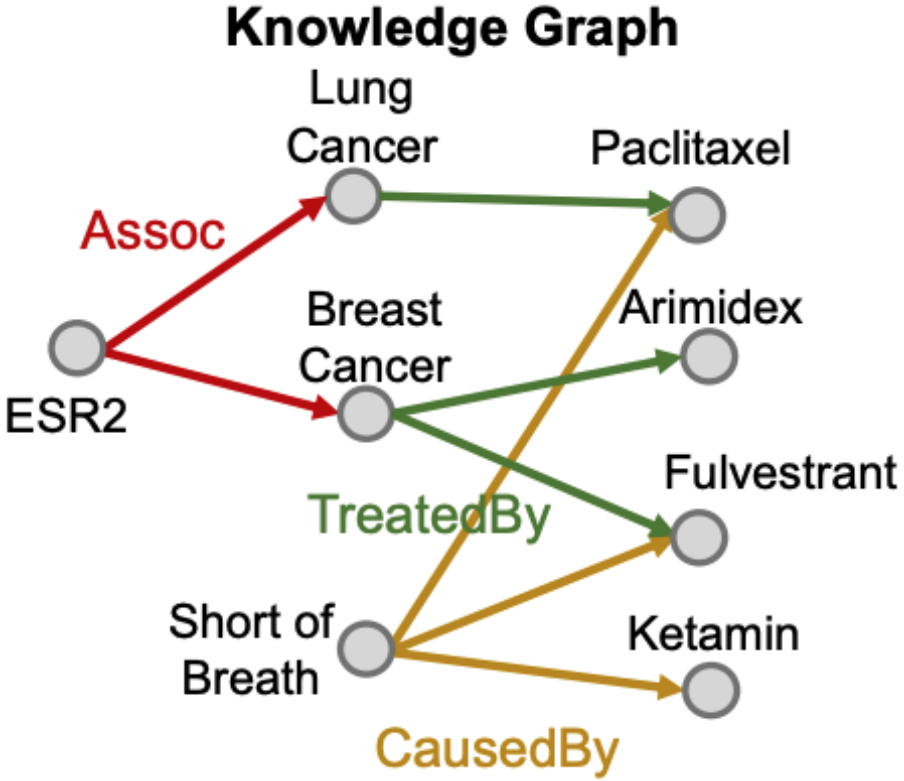
Since we are considering an intersection, we require that Fulvestrant is the entity ending the query paths at the preceding level in the query.

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Generating sample queries on a KG

- How to do this systematically?



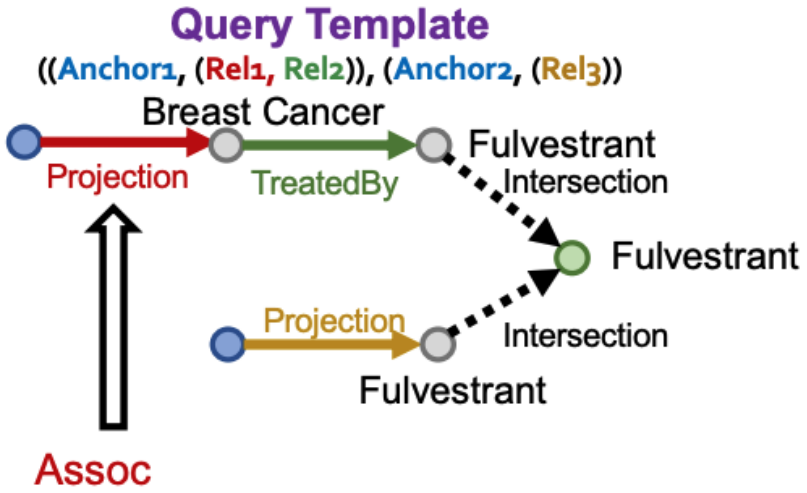


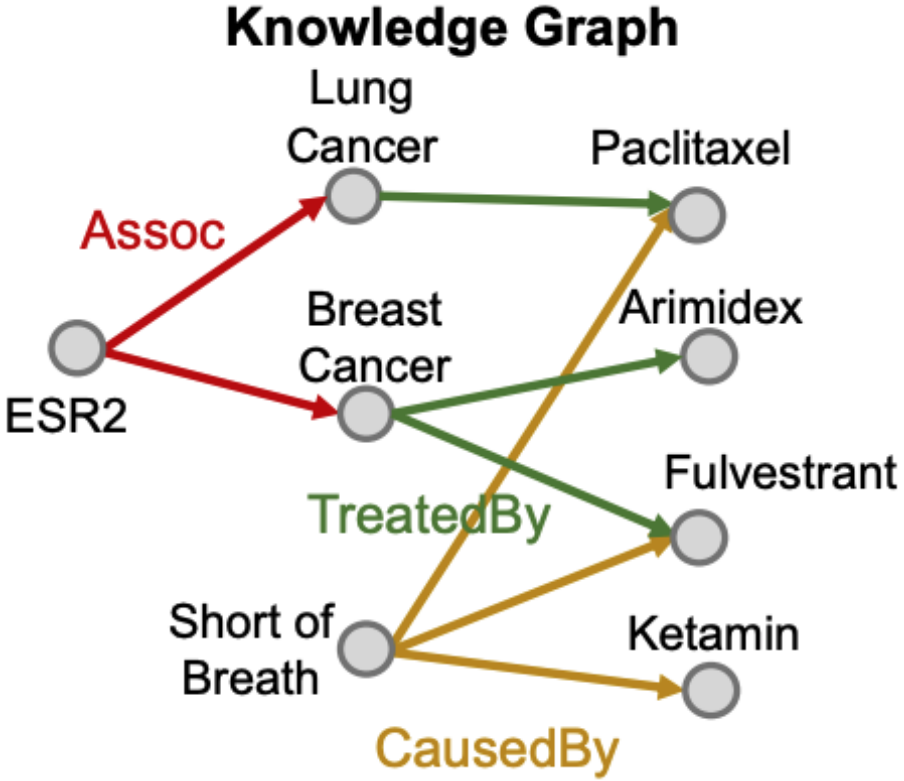
Pick on of the branches (top). Instantiate the **projection edge** in the template by randomly sampling one relation associated with the current entity, Fulvestrant. For example, we might select **TreatedBy** and then check what entities are connected to Fulvestrant with **TreatedBy**: {Breast Cancer}

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Generating sample queries on a KG

- How to do this systematically?



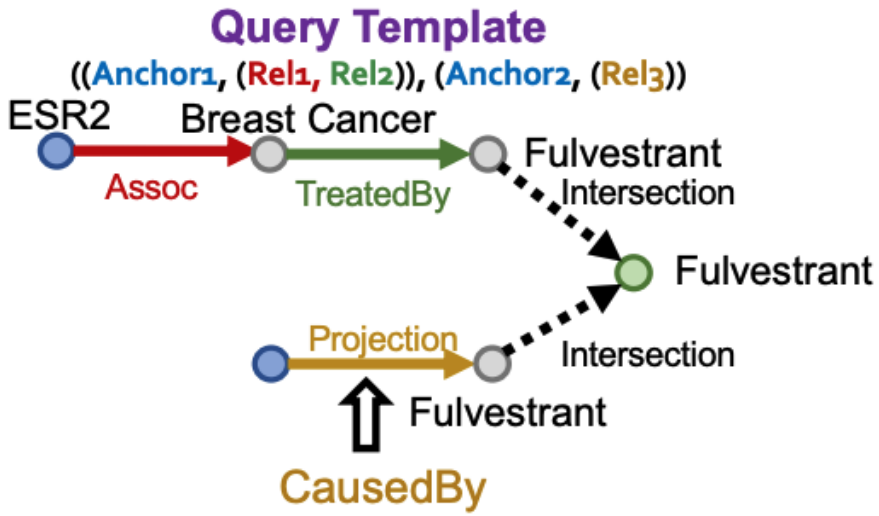


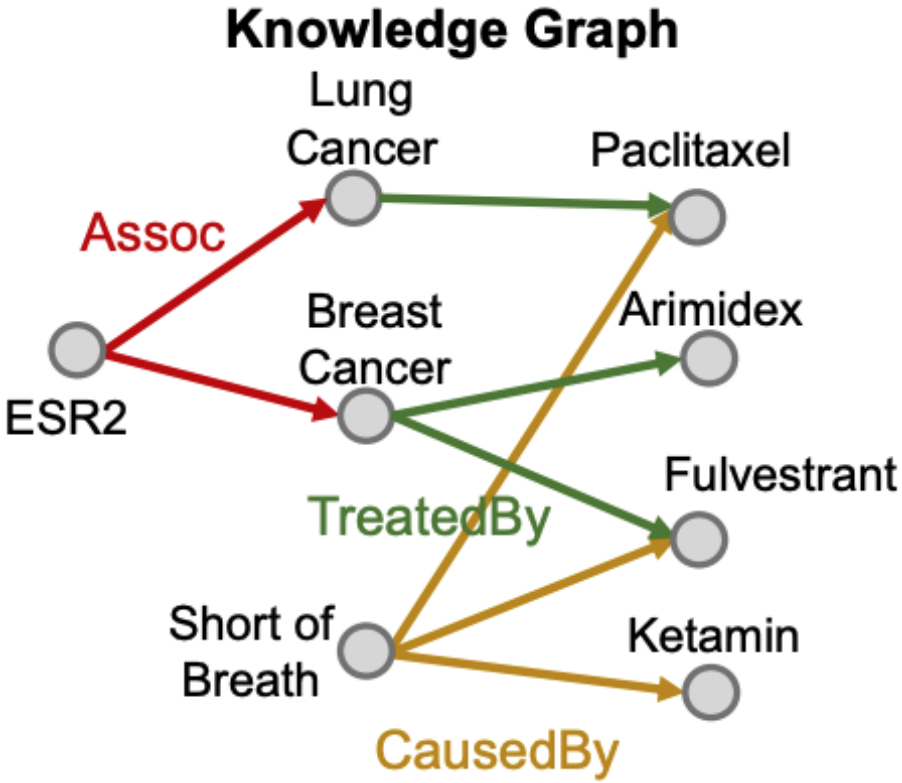
Staying on the top branch, instantiate the next **projection edge** by randomly sampling one relation associated with Breast Cancer. For example, we might select **Assoc** and then check what entities are connected to Breast Cancer with **Assoc**: {ESR2}

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Generating sample queries on a KG

- How to do this systematically?



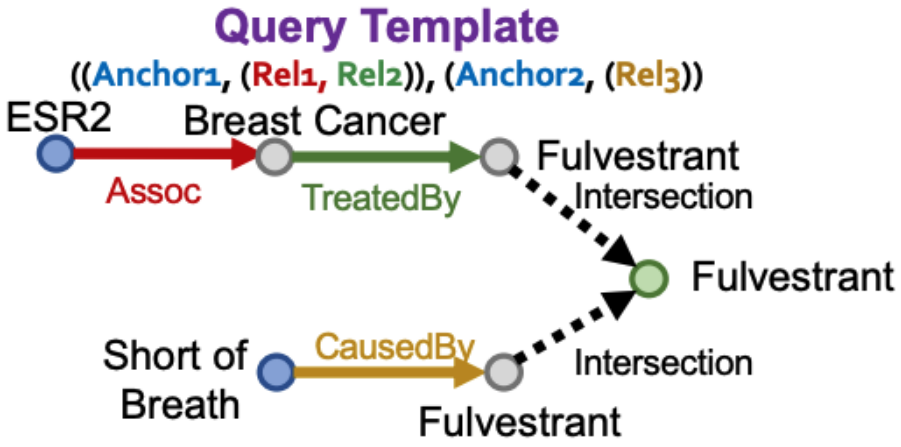


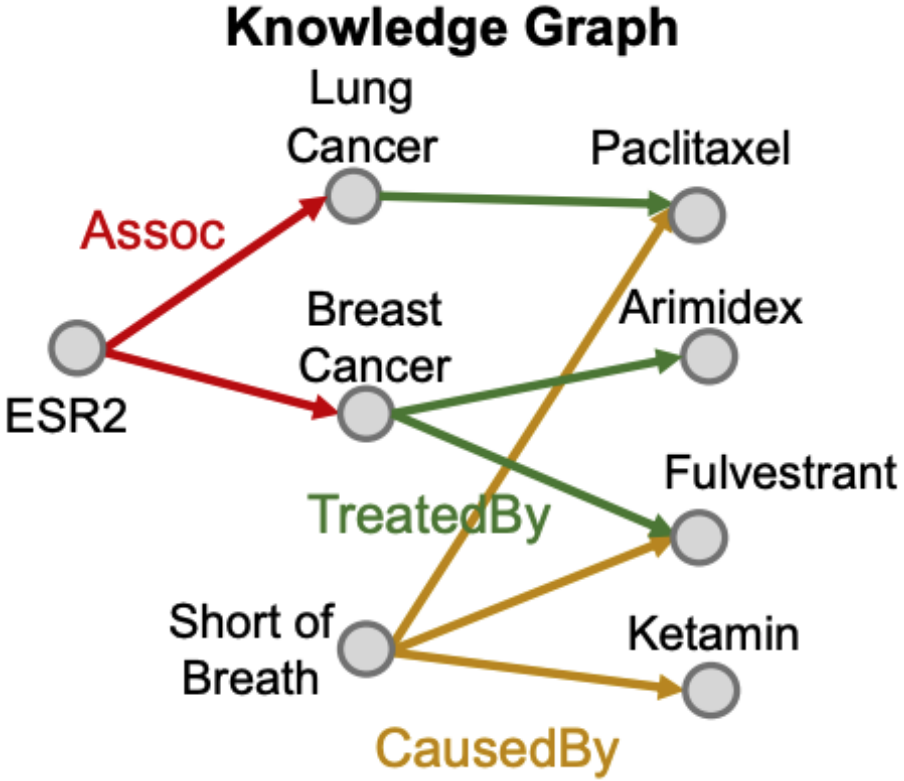
Now we move to the next (bottom) branch. We instantiate the **projection edge** by randomly sampling one relation associated with Fulvestrant. For example, we might select **CausedBy** and then check what entities are connected to Fulvestrant with **CausedBy**: {Short of Breath}

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Generating sample queries on a KG

- How to do this systematically?





Finally, we select the last entity from {Short of Breath}

Image credit: Jure Leskovec, http://cs224w.stanford.edu

Generating sample queries on a KG

- We now have a fully instantiated query

  [(e:ESR2, (r:Assoc, r:TreatedBy)), (e:Short of Breath, (r:CausedBy))]
- We can obtain the "known" answers by traversing the KG – this will NOT account for incompleteness
- We can then randomly select negative answers that are not in the known answers (there is the chance of false negatives)

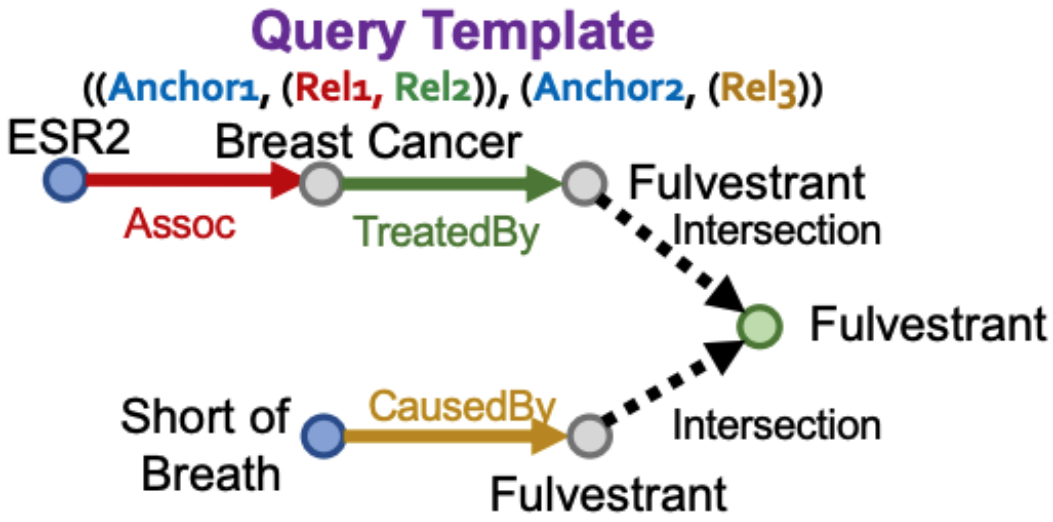


Image credit: Jure Leskovec, http://cs224w.stanford.edu